Real-Time Reliability of First-Release Nonfarm Payroll Estimates: Short- and Long-Horizon Evidence from CES Revisions, 2000–2024

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Abstract

By conceptualizing payroll revision magnitude as a regime-dependent stochastic process rather than classical measurement error, this paper evaluates the real-time reliability of U.S. nonfarm payroll estimates over 2000–2024. Drawing on a vintage-consistent panel linking initial CES "first prints" to their latest-available QCEW-benchmarked counterparts, the analysis documents a heavy-tailed revision distribution in which tail events exceeding 200,000 jobs occur not as rare anomalies but as recurrent features of the measurement process, with exceedance probabilities reaching 5 percent under baseline conditions and 20 percent during COVID. An Interrupted Time Series design centered on COVID-19 onset identifies a statistically significant upward level shift of 168,000 jobs in revision magnitude that persists 57 months post-shock, accompanied by a 39-percentage-point jump in tail-event probability that remains structurally elevated nearly five years later, indicating a durable operational break rather than transitory volatility. Revision variance displays sharp counter-cyclical behavior: initial estimates systematically overstate employment at cyclical turning points due to breakdown of backward-looking birth-death adjustments when firm dynamics depart from historical precedent. Quantile-focused machine learning models estimated on strictly real-time information sets improve average calibration but systematically fail to forecast extreme outcomes during structural strain, with tail misses clustering around the 2008–09 financial crisis and 2020–22 pandemic window, underscoring fundamental limits when survey infrastructure degrades. Overall, the findings demonstrate that the informational content of the first payroll print is regime contingent—static confidence intervals assuming constant variance understate uncertainty precisely when it matters most, and credible real-time inference requires explicit adjustment for time-varying, state-dependent measurement risk rather than reliance on fixed margins of error applied uniformly across the business cycle.

1. Introduction

Released on August 1, 2025, the Bureau of Labor Statistics Employment Situation report for July immediately drew attention beyond its headline estimate of a 73,000 increase in nonfarm payrolls, as concurrently published revisions reduced the combined May and June totals by 258,000 jobs (Bronars, 2025). Concentration of market focus on these downward adjustments reflected their scale, which ranked among the largest two-month revisions observed in the post-pandemic period and triggered pronounced volatility across financial markets alongside renewed scrutiny of the bureau's estimation procedures. Subsequent removal of BLS Commissioner Erika McEntarfer intensified debate over the origin of the discrepancy, with competing interpretations attributing the episode either to political dissatisfaction with an emerging signal of labor market deceleration or to erosion within the institutional machinery responsible for labor measurement (Ackerman & Bogage, 2025).

Embedded within this episode lies a broader analytical tension central to contemporary macroeconomic assessment. Emphasis within policy discourse frequently gravitates toward the accuracy of a single release, yet the more consequential issue concerns the stability and reliability of the measurement process itself. Determining whether the August 2025 revisions constitute an isolated disturbance or evidence of deeper structural discontinuities requires displacement of political framing in favor of systematic analysis of revision behavior over time. Evidence presented here indicates that volatility observed in late 2025 aligns with a post-2020 regime shift characterized by a materially higher variance in initial payroll estimates, suggesting a persistent alteration in the informational content of early labor market signals rather than a singular statistical aberration.

1.1 The Economic Function of Preliminary Estimates

Functioning as a central input into high-frequency capital allocation and the calibration of monetary policy, the preliminary estimate of nonfarm payrolls, or the "first print," exerts influence far in excess of its provisional character. Financial markets and the Federal Reserve respond to these initial signals in real time, even as the fully benchmarked series constructed from comprehensive Unemployment Insurance tax records remains unavailable for extended periods, often stretching into years. Resulting from this temporal gap is a persistent information asymmetry in which consequential decisions are rendered under conditions of pronounced measurement uncertainty rather than statistical finality.

Within traditional macroeconomic practice, such uncertainty has generally been treated as classical measurement error, understood as random, mean-zero variation that attenuates as additional data accrue (Mankiw & Shapiro, 1986). Framed in this manner, early estimates are presumed to retain unbiased predictive content, with subsequent revisions interpreted as largely mechanical accounting exercises. Emerging empirical work has increasingly complicated this assumption. Findings presented by Cajner et al. (2019) indicate that structural transformations in labor market dynamics can weaken the predictive power of official survey-based measures relative to certain private-sector alternatives, particularly near cyclical inflection points. Complementary analysis from the Federal Reserve Bank of Philadelphia (2023) further demonstrates that so-called early benchmark revisions exhibit systematic forecasting ability with respect to final BLS corrections, suggesting that revision behavior embeds structural information rather than reflecting transitory statistical noise.

1.2 Research Objectives and Methodology

By conceptualizing payroll revision magnitude as a regime-dependent stochastic process, the study advances a formal framework for evaluating the reliability of employment measurement in real time. Drawing on a twenty-five-year panel of vintage-based payroll revisions spanning 2000–2024, the empirical design addresses three interrelated questions concerning distributional stability, regime dependence, and the feasibility of real-time risk assessment.[1]

[1] This analysis is limited to the period 2000–2024 because the 2025 nonfarm payroll series has been subject to a major QCEW-based benchmark revision finalized in the January 2026 Current Employment Statistics (CES) report, which downwardly revises total nonfarm employment by roughly 862,000 jobs (−0.5 percent) at the March 2025 level (Mutikani, 2026). The 2025 series is further complicated by the fact that the final 2025 QCEW counts will not be published until the 4th-quarter 2025 QCEW news release on June 2, 2026 (U.S.

Analytical progression unfolds in three stages. Section 4 introduces exploratory distributional evidence showing that, although the median revision remains relatively contained at approximately 53,000 jobs, the surrounding distribution displays pronounced heavy-tailed characteristics. Section 5 then applies Interrupted Time Series and event-study methodologies to isolate the effects of the COVID-19 shock, identifying a statistically significant and persistent upward shift in revision volatility. Section 6 concludes the empirical sequence by assessing quantile regression performance in forecasting revision risk, revealing systematic failures of conventional calibration techniques to anticipate tail outcomes during episodes of structural strain.

1.3 Summary of Findings

Conditioned on the prevailing economic regime, the informational reliability of initial payroll estimates emerges as sharply uneven across the business cycle. Empirical results indicate that revision variance behaves counter-cyclically, with early estimates displaying a systematic propensity to overstate employment levels at the onset of downturns, a pattern commonly characterized as turning point bias. Compounding this asymmetry, the COVID-19 pandemic introduced a durable structural break in measurement volatility, with dispersion failing to revert fully to its pre-2020 baseline even as headline labor conditions normalized. Further erosion appears in the domain of predictability, where models designed to forecast revision risk deteriorate precisely during episodes of elevated stress, reflecting non-stationarity in the revision-generating process under shock conditions. Interpretation of real-time labor market data therefore necessitates dynamic, regime-contingent adjustments that explicitly account for time-varying measurement uncertainty rather than reliance on static assumptions of stability.

Bureau of Labor Statistics, 2025b). Until that full QCEW-2025 dataset is available and incorporated into the broader research community's work, the 2025 employment path remains in flux for analytical applications and is therefore excluded from the estimation sample.

2. Institutional Background and Related Literature

To clarify the magnitude and behavior of payroll revisions, an initial examination of how employment estimates are constructed, the intervals at which they are revised, and the institutional meaning attached to so-called final values is required. This section details the dual-track measurement architecture employed by the Bureau of Labor Statistics, explicates the sequencing and mechanics of the revision cycle, and situates the present analysis within the broader literature on real-time macroeconomic data uncertainty. Framed in this manner, revision magnitude is treated neither as a residual nuisance nor as a purely technical artifact, instead emerging as a regime-dependent characteristic of the information environment shaped jointly by evolving economic structure and the operational constraints embedded within the measurement apparatus itself.

2.1 The Institutional Production of Nonfarm Payroll Estimates

Structured to mediate the tension between reporting immediacy and statistical completeness, monthly nonfarm payroll estimation at the Bureau of Labor Statistics operates through a deliberately layered dual-track system. Initial figures are produced within the Current Employment Statistics program, which surveys approximately 119,000 businesses and government entities and captures roughly one-third of total nonfarm employment. Even with a long record of institutional stability, this survey-based framework confronts a structural limitation rooted in its design, as real-time identification of firm entry and exit remains infeasible. Mitigation occurs through deployment of a net birth-death model, implemented using an ARIMA-based statistical procedure that infers establishment formation and closure from historical employment dynamics, as documented by the Bureau of Labor Statistics (2025a). Emerging from this architecture, the first-Friday payroll release reflects a composite of reported survey responses and model-derived adjustments, a configuration that becomes increasingly exposed during regime shifts when contemporary firm behavior departs from historical precedent.

Established only after a substantial reporting delay, the benchmark employment count is derived from the Quarterly Census of Employment and Wages, which aggregates unemployment insurance tax records from more than eleven million establishments nationwide. Coverage exceeds ninety-five percent of U.S. employment, conferring a degree of comprehensiveness unavailable to contemporaneous surveys, while publication occurs with an approximate five-month lag. Separation between the timely yet partially imputed CES estimates and the delayed but near-census QCEW counts forms the institutional foundation of payroll revisions. Revision magnitude examined here arises directly from this temporal and methodological gap, reflecting the interaction between measurement architecture and evolving economic conditions rather than transitory statistical noise.

2.2 Revision Timing, Benchmarking, and the Meaning of "Latest-Available" Data

Revisions to payroll data unfold along two distinct temporal horizons, encompassing both near-term survey updating and longer-term institutional reconciliation. Short-horizon revisions occur in the two months following the initial release as delayed survey responses are incorporated into the sample framework. Under standard release conventions, a January reference month first appears in February, is revised in March, and receives a third print in April. Adjustments at this stage primarily reflect correction of sampling variability and non-response bias, with limited exposure to deeper sources of structural mismeasurement.

Long-horizon revisions proceed on a slower annual cadence through benchmarking exercises that align the sample-based Current Employment Statistics series with the administrative universe captured in the Quarterly Census of Employment and Wages. Completion of this process each February addresses cumulative drift arising from the birth–death model and other persistent estimation errors that accumulate over time. For purposes of the present analysis, the "latest-available" value is defined as the data vintage observed in December 2024, treated as the operative historical record rather than an immutable statistical endpoint. This terminal vintage reflects incorporation of all available administrative benchmarks and methodological refinements consistent with established approaches to real-time data uncertainty in the macroeconomic literature (Croushore & Stark, 2001).

2.3 Conceptual Framework: Reliability, Uncertainty, and Real-Time Inference

Situated within this institutional architecture, revision magnitude functions as an empirical proxy for measurement reliability, defined less by the elimination of error than by the stability of its distributional properties over time. Reliability, in this framing, hinges on whether revisions approximate white noise, characterized by unpredictability, mean-zero behavior, and constant variance, or whether they instead display colored noise features such as serial correlation, recession-contingent bias, or fat-tailed dispersion during periods of systemic stress. Emergence of such structure materially degrades the informational content of the first print, with direct implications for real-time inference. Under conditions of stable variance, decision-makers can reasonably apply a fixed confidence interval across releases. Once variance becomes regime dependent, expanding sharply during episodes such as pandemics or financial crises, reliance on static uncertainty bands becomes misleading. Interpretation of payroll signals therefore depends on economic state, rendering reliability a dynamic attribute of the information environment rather than a fixed property of the Bureau of Labor Statistics methodology.

2.4 Related Literature and Positioning

Extending across several adjacent literatures, scholarship on payroll revision behavior coalesces around three interrelated lines of inquiry: the foundational dispute over whether revisions convey informational content or merely correct transitory noise, the use of real-time data vintages in the evaluation of policy decisions, and the measurement challenges that emerge as labor markets approach cyclical inflection points. This subsection examines each strand sequentially before situating the present analysis within this broader research agenda, emphasizing points of convergence and unresolved tensions that motivate a regime-sensitive approach to revision dynamics.

2.4.1 The News vs. Noise Debate

Originating with the seminal contribution of Mankiw and Shapiro (1986), the literature on data revisions crystallized around the distinction between revisions as carriers of news versus corrections of noise. Their analysis of GNP revisions advanced the view that initial estimates functioned as rational forecasts of final values, thereby characterizing revisions as largely unpredictable arrivals of new information. Subsequent empirical work drawing on richer real-time datasets has increasingly complicated this interpretation. Evidence presented by Aruoba (2008) demonstrates that revisions to core macroeconomic indicators frequently display non-zero means and measurable predictability, patterns inconsistent with rational expectations and suggestive of systematic noise embedded within early estimates. Parallel findings emerge in the cross-country context, where Faust et al. (2005) document predictable components in GDP revisions across G-7 economies, attributing such structure to statistical smoothing practices that leave residual information in the error process. Extending this line of inquiry, Fixler et al. (2024) examine U.S. GDP and GDI revisions and identify clear regime-dependent behavior, with revision dynamics intensifying around business cycle turning points, thereby reinforcing the view that revision processes themselves vary with underlying economic conditions.

2.4.2 Real-Time Data and Policy Evaluation

Recognition that data revisions exert material influence on inference has reshaped the evaluation of macroeconomic policy across multiple domains. Early evidence provided by Orphanides (2001) demonstrated that monetary policy rules yield markedly different conclusions when assessed using real-time data rather than ex post revised series, revealing that several policy errors attributed to the Federal Reserve during the 1970s were closely linked to contemporaneous mismeasurement of the output gap. Institutionalizing this insight, Croushore and Stark (2001) developed the foundational infrastructure for real-time econometric analysis through the construction of vintage-dated datasets that replicate the informational environment confronting policymakers at each historical decision point. Subsequent contributions by Koenig et al. (2003) sharpened the methodological critique by illustrating how forecasts evaluated with revised data can materially overstate real-time performance, particularly when analysts inadvertently combine inconsistent data vintages.

Extending this framework toward operational policy analysis, Giannone et al. (2008) advanced nowcasting methodologies that explicitly model the staggered arrival of economic indicators, reframing real-time inference as a problem of systematically missing information. Their approach underscored that uncertainty in policy environments arises as much from data availability and revision dynamics as from structural model error. Within this literature, revisions cease to be a secondary statistical concern and instead become central to understanding why policy assessments diverge so sharply between real-time and retrospective perspectives, especially during periods of rapid economic transition.

2.4.3 Labor Market Measurement and Predictability

Situated within labor economics, a growing body of research has concentrated on the informational content of alternative data sources and the behavior of revisions around cyclical turning points. Using high-frequency private payroll data from ADP, Cajner et al. (2019) show that official CES estimates tend to lag underlying labor market conditions at inflection points, with blended public–private measures reducing mean squared revision error by isolating structural noise embedded in the official release. Parallel evidence from the Federal Reserve Bank of Philadelphia (2023) demonstrates that so-called early benchmarks, constructed by aggregating state-level QCEW data, possess meaningful predictive power for subsequent annual BLS revisions. Additional distortions arise from the mechanics of seasonal adjustment, as Wright (2013) documents that standard filters struggle to differentiate permanent level shifts from transient seasonal fluctuations during recessions, thereby amplifying measurement error. Reinforcing this pattern, Quinlan and Pinheiro (2025) find that employment forecast errors widen systematically during downturns, consistent with a counter-cyclical expansion in measurement uncertainty rather than deterioration in forecaster skill alone.

2.4.4 Positioning of This Study

Distinguished by a reorientation of analytical emphasis, this paper shifts attention away from point prediction and toward explicit quantification of revision uncertainty. Existing research has largely focused on whether the direction or magnitude of payroll revisions can be anticipated, while the present analysis instead examines whether the uncertainty surrounding those revisions can be characterized contemporaneously. Contribution unfolds across three dimensions. First, the study develops a comprehensive distributional account of payroll revisions spanning twenty-five years, treating the COVID-19 episode as a natural experiment in extreme measurement stress. Second, turning point bias is formalized as a regime-dependent feature of the revision-generating process, rooted in structural conditions rather than transient sampling disturbances. Third, feasibility of real-time uncertainty assessment is evaluated through application of machine learning methods, demonstrating that conventional calibration techniques deteriorate precisely during periods when tail risk becomes most acute. By modeling the conditional distribution of revisions with particular emphasis on extreme outcomes, the analysis reframes the first payroll print as a probabilistic object that demands risk-based interpretation rather than acceptance as a definitive scalar signal.

2.5 Scope and Interpretive Boundaries

Delimiting the interpretive scope of the analysis is essential to prevent mischaracterization of its conclusions. Primary emphasis rests on evaluating the reliability of the public signal, not on assessing the competence or integrity of the public institution responsible for its production. Structural disruptions such as the COVID-19 pandemic impose severe constraints on any statistical system, and evidence of diminished reliability reflects the complexity of the estimation environment rather than allegations of malpractice or intentional distortion. Treatment of the "latest-available" benchmark likewise warrants qualification. This reference series functions here as the operative standard against which revisions are measured, even though it remains subject to residual administrative error and the possibility of future methodological refinement. Finally, orientation of the study is explicitly retrospective. Although feasibility of real-time risk modeling is examined, no claim is made regarding the provision of an operational forecasting tool for future monthly releases. Analytical intent is diagnostic and descriptive, aimed at clarifying the structure of uncertainty embedded in real-time payroll data rather than prescribing policy or procedural remedies.

3. Data and Construction

Establishing the empirical foundation of the study requires precise specification of data inputs, sample boundaries, and the construction of all revision measures and regime classifications employed throughout the analysis. Empirical work is conducted on a single monthly panel spanning January 2000 through December 2024, yielding 300 observations. All construction choices are mechanical, fully specified ex ante, and invariant to subsequent results. Consistent with this objective, the section confines itself to definitional and procedural exposition, with no presentation of distributional statistics, timing diagnostics, or predictive modeling. Progression across empirical sections follows a deliberate sequence from description to diagnosis to modeling, anchored throughout by a unified and stable data environment constructed here.

3.1 Data Sources, Observation Unit, and Sample Definition

Defined at the monthly frequency, the unit of observation corresponds to calendar month t, indexed in YYYY-MM format and structured to capture sequential information updates for each reference period. For every month, the dataset records three release-vintage estimates of the over-the-month change in total nonfarm payroll employment, labeled `first_estimate`, `second_estimate`, and `third_estimate`. These vintages are drawn directly from Current Employment Statistics revision materials published by the Bureau of Labor Statistics, which report seasonally adjusted employment change figures as released in the first, second, and third Employment Situation reports associated with each reference month (Bureau of Labor

Statistics, 2024).

Complementing the release-vintage series, the dataset includes a latest-available reference measure of total nonfarm payroll employment levels, `PAYEMS`, along with its corresponding month-over-month change, `PAYEMS_CHG`, obtained from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis. Construction of `PAYEMS_CHG` follows mechanically as the first difference of `PAYEMS`. All variables capturing employment levels or changes are expressed in thousands of jobs. Sample inclusion is restricted to months for which all required release-vintage estimates and reference series are jointly available to construct the revision measures defined below, producing uninterrupted monthly coverage across the stated horizon.

3.2 Construction of Revision Measures

Distinction within the paper is drawn between short-horizon revisions arising inside the standard release cycle and a long-horizon revision defined relative to the latest-available benchmark. Short-horizon revisions are constructed mechanically as differences across successive release vintages for a common reference month. The first-to-second revision is defined as

$$\texttt{revision\_1\_2}_t = \texttt{second\_estimate}_t - \texttt{first\_estimate}_t,$$

the second-to-third revision as

$$\texttt{revision\_2\_3}_t = \texttt{third\_estimate}_t - \texttt{second\_estimate}_t,$$

and the cumulative first-to-third revision as

$$\texttt{revision\_1\_3}_t = \texttt{third\_estimate}_t - \texttt{first\_estimate}_t.$$

Each measure captures incremental updating of the initial payroll signal as additional survey information enters across successive releases.

Defined on a longer horizon, the benchmark revision compares the first-release estimate to the latest-available month-over-month change in payroll employment:

$$\texttt{net\_long\_revision}_t = \texttt{PAYEMS\_CHG}_t - \texttt{first\_estimate}_t,$$

with the corresponding unsigned magnitude given by

$$\texttt{abs\_net\_revision}_t = |\texttt{net\_long\_revision}_t|.$$

Interpretation assigns directional drift to the signed quantity `net_long_revision`, while `abs_net_revision` represents the size of the long-horizon adjustment and serves as the primary reliability magnitude object analyzed in Sections 4 through 6.

Internal coherence of the construction is ensured by two accounting identities that follow directly from the definitions. Exact decomposition holds between cumulative and incremental short-horizon revisions:

$$\texttt{revision\_1\_3}_t = \texttt{revision\_1\_2}_t + \texttt{revision\_2\_3}_t.$$

Consistency between long-horizon revisions and benchmark data is likewise guaranteed by construction:

$$\texttt{net\_long\_revision}_t = \texttt{PAYEMS\_CHG}_t - \texttt{first\_estimate}_t.$$

Both identities are satisfied exactly for every observation in the sample.

3.3 Regime Labels and Derived Indicators

Supporting stratified description and conditional diagnostics in later sections, several time-based labels are incorporated into the dataset. Monthly recession status is captured by `USREC`, a binary indicator denoting NBER-defined recessions, retrieved from the Federal Reserve Economic Data (FRED) database. Broader historical context is provided by `macro_regime`, a coarse partition of the sample into three macroeconomic eras: Pre-GFC Expansion spanning 2000-01 through 2007-12, Post-GFC / ZLB Era spanning 2008-01 through 2019-12, and

COVID & Aftermath spanning 2020-01 through 2024-12. Political context is recorded through `administration`, identifying the presidential administration in office during month t (Clinton, G.W. Bush, Obama, Trump, Biden), and `party`, denoting the party affiliation of the sitting president (Democratic, Republican). Assignment of all regime indicators follows calendar-month rules applied mechanically, with January of an inauguration year attributed to the outgoing administration. These labels function exclusively as descriptive partitions rather than causal treatments.

Beyond baseline classifications, additional derived indicators are constructed to support later analysis and are defined here to ensure consistency across subsequent sections. Tail-event flags are generated by applying fixed thresholds to `abs_net_revision`, with illustrative cutoffs at 100,000, 200,000, and 300,000 jobs. Rolling-window summaries and lagged revision-history features are likewise constructed using information strictly available prior to the reference month when employed in predictive settings. All such derived objects are specified at this stage to avoid redefinition in Sections 4 through 6 and to preserve a clear separation between data construction and empirical analysis.

3.4 Data Integrity and Reproducibility

Preservation of a single and internally consistent measurement environment across all empirical sections requires that the dataset undergo a sequence of integrity checks prior to analysis. Verification confirms continuous monthly coverage from 2000-01 through 2024-12 without duplication, uniform units across all payroll change and revision measures, and exact satisfaction of the two accounting identities linking short-horizon revisions to the long-horizon definition. This same monthly panel underlies the exploratory distributional analysis in Section 4, the quasi-causal timing and regime diagnostics in Section 5, and the predictive modeling exercises in Section 6, ensuring that observed differences across sections arise from analytical design rather than shifts in sample composition or variable construction. All data sources are publicly available, and the final constructed dataset contains no proprietary or restricted information. Replication is facilitated through availability of both the constructed dataset and the code used to generate all revision measures and regime labels upon request.

Section 4 builds directly on this common empirical foundation by characterizing the unconditional and regime-conditional distributions of the constructed revision measures. Resulting regularities motivate the timing-based diagnostics developed in Section 5 and inform the uncertainty-focused modeling framework introduced in Section 6.

4. Exploratory Distributional Evidence

Having established the data architecture and construction protocols in Section 3, attention now turns to the empirical characterization of revision behavior across the full sample period. This section proceeds through a systematic distributional analysis designed to uncover stylized facts regarding the magnitude, symmetry, regime dependence, temporal variation, and correlation structure of payroll revisions. Organization follows five subsections, each targeting a distinct dimension of the revision-generating process. Section 4.1 examines unconditional revision behavior, establishing baseline distributional properties and documenting the pronounced heavy-tailed character of long-horizon adjustments. Section 4.2 isolates tail risk, quantifying exceedance frequencies and demonstrating the disproportionate contribution of the COVID-19 period to extreme revisions. Section 4.3 stratifies revisions by economic regime, revealing systematic patterns tied to recession episodes, macroeconomic eras, and political administrations. Section 4.4 traces temporal evolution, identifying periods of elevated volatility and persistent directional drift that extend well beyond the immediate crisis window. Section 4.5 investigates correlation structure, assessing the informational linkages between first-release payroll estimates and subsequent revision outcomes. Collectively, these subsections yield a set of empirical regularities regarding the magnitude, structure, and regime-dependence of payroll revision risk. Presentation throughout remains strictly descriptive, reserving causal interpretation and formal modeling for subsequent sections.

4.1 Unconditional Revision Behavior and Distributional Properties

Establishing the empirical foundation for subsequent regime-specific and temporal analyses requires first documenting the unconditional statistical properties of payroll revisions over the full 2000–2024 sample. The analysis distinguishes explicitly between short-horizon revisions, which reflect information arriving within the first few subsequent releases, and long-horizon outcomes, which capture uncertainty that resolves only through later benchmarking and extended revision cycles. Presentation throughout this subsection remains strictly descriptive, aimed at characterizing the magnitude, dispersion, and asymmetry of real-time measurement uncertainty embedded in first-release nonfarm payroll estimates without imposing causal structure or normative judgment.

Two interpretive choices guide the analysis. First, summary statements emphasize medians, interquartile ranges, and upper-tail quantiles rather than means, as revision distributions are not well approximated by Gaussian behavior. Second, signed and absolute revisions are treated as distinct statistical objects. Signed revisions describe directional drift, while absolute revisions measure the magnitude of uncertainty confronting real-time users prior to the arrival of later information.

4.1.1 Typical Magnitudes and Central Tendency Across Revision Horizons

Across the full sample, short-horizon revisions exhibit central tendencies close to zero. Typical updates between the first and subsequent releases are small relative to the headline monthly payroll change, and the interquartile ranges are correspondingly narrow. This pattern is consistent with a revision process in which early updates primarily reflect incremental survey response accumulation and routine processing adjustments rather than wholesale re-estimation of labor market conditions.

Long-horizon outcomes behave differently. Even when the median net long-horizon revision remains near zero, the distribution is substantially wider than the short-horizon distributions. The interquartile range expands markedly, and dispersion increases well beyond what is visible in the first few releases. The key implication is that early revisions capture only a fraction of total measurement uncertainty. A meaningful share of uncertainty remains unresolved until later benchmarking and extended revision windows, well beyond the horizon typically emphasized in real-time interpretation.

4.1.2 Absolute Revisions as the Relevant Object for Measurement Uncertainty

Signed summaries can be misleading in revision data because positive and negative errors may offset one another. This limitation becomes immediately apparent when revisions are evaluated in absolute terms. For net long-horizon revisions, the median absolute revision is large relative to the median signed revision. A near-zero signed median therefore does not imply low uncertainty; it indicates only that the distribution is centered near zero, not that outcomes are tightly concentrated around that center.

Absolute measures also clarify the role of tail outcomes. Mean absolute revisions exceed median absolute revisions by a wide margin in both the full sample and the non-COVID subsample, reflecting the influence of infrequent but very large revision episodes. This divergence signals that "average" months and "tail" months occupy fundamentally different parts of the distribution.

4.1.3 Long-Horizon Dispersion and Upper-Tail Exposure

Long-horizon net revisions exhibit materially greater dispersion than short-horizon revisions. Even outside the pandemic window, the interquartile range for the net long-horizon revision is on the order of 100 thousand jobs, and upper-tail quantiles rise rapidly relative to the IQR. These features establish a basic stylized fact: the revision process contains heavy-tailed risk that is economically meaningful even under normal conditions.

This unconditional evidence indicates that large revisions are not confined to extraordinary periods. Rather, tail exposure is an inherent feature of long-horizon payroll measurement. To maintain a clear separation between unconditional properties and regime-specific amplification, explicit probability statements regarding tail exceedances are deferred to Section 4.2.

4.1.4 The Pandemic as a Distinct High-Variance Case Study

The COVID-only subsample exhibits behavior characteristic of a distinct variance regime. Dispersion in both signed and absolute long-horizon revisions expands sharply, upper-tail quantiles move to an entirely different scale, and extreme realizations become common within a short window. These outcomes reflect simultaneous disruptions to data collection, classification, and underlying labor market dynamics.

Because this regime is not representative of baseline measurement uncertainty, COVID months are treated as a high-variance case study rather than pooled with non-COVID months for statements about typical reliability. This separation ensures that unconditional summaries are not dominated by a short but extraordinary episode, while still preserving COVID as an analytically informative stress test of the revision process.

Table 1: Summary statistics for long-horizon revisions (thousands of jobs)

	Signed net long-horizon revision			Absolute net revision	
Sample	Median	IQR	P95	Median	P95
Full sample (2000–2024)	4.5	105.0	141.25	53.0	209.4
Non-COVID sample	3.5	104.0	135.85	52.5	201.2
COVID-only sample	39.0	234.25	246.35	138.0	513.3

4.1.5 Interpretation

Unconditionally, short-horizon revisions are typically small and centered near zero, while long-horizon outcomes display substantially greater dispersion. Absolute revisions reveal that the economically relevant object is uncertainty magnitude rather than signed central tendency. Finally, the pandemic window constitutes a distinct high-variance environment that must be handled separately from baseline periods. The substantial dispersion documented here raises the question of how frequently revisions of economically meaningful size actually occur.

4.2 Heavy Tails and Asymmetry in Long-Horizon Payroll Revisions

Establishing that long-horizon revisions exhibit wide dispersion leaves open the question of how frequently that dispersion manifests as economically salient tail events. Exceedance probabilities quantify the share of months in which the absolute net long-horizon revision exceeds fixed thresholds, providing a direct measure of tail risk. Distributional asymmetry is summarized using skewness and kurtosis to clarify why signed moments provide unstable characterizations in high-variance environments.

4.2.1 Baseline Tail Behavior Outside COVID

Even outside the COVID window, tail exposure in long-horizon payroll revisions is substantial. Approximately one-fifth of non-COVID months experience an absolute net revision of at least 100 thousand jobs. Revisions exceeding 200 thousand jobs occur in roughly one out of twenty months, and exceedances above 300 thousand jobs, while infrequent, are not vanishingly rare.

These frequencies are inconsistent with a thin-tailed or approximately normal error process. Under such a process, events of this magnitude would be extremely unlikely. Instead, the observed exceedance rates imply that large revisions are a recurring feature of the measurement process rather than statistical anomalies. In practical terms, revisions large enough to alter real-time narratives about labor market momentum should be regarded as plausible outcomes over multi-year horizons, even during periods that appear calm in aggregate.

4.2.2 COVID as a Tail-Amplification Regime

The COVID-only window exhibits a sharply different tail profile. Exceedance probabilities rise at every threshold, and the degree of amplification increases monotonically as the threshold moves deeper into the tail. Relative to the non-COVID sample, the probability of exceeding 100 thousand jobs is approximately 2.7 times larger, the probability of exceeding 200 thousand jobs is nearly 4 times larger, and the probability of exceeding 300 thousand jobs is more than

7 times larger.

COVID does not simply scale up overall variance uniformly. Instead, it disproportionately inflates extreme outcomes, producing a nonlinear thickening of the tails. The result is a regime in which very large revisions become not merely possible but common within a short time window.

Table 2: Exceedance probabilities for absolute net long-horizon revisions

Sample	$\Pr(\lvert revision \rvert \geq 100K)$	$\Pr(\lvert revision \rvert \geq 200K)$	$\Pr(\lvert revision \rvert \geq 300K)$
Full sample (2000–2024)	0.233	0.057	0.017
Non-COVID sample	0.221	0.052	0.014
COVID-only sample	0.600	0.200	0.100

4.2.3 Asymmetry and the Limits of Signed Moments

Heavy tails interact with asymmetry in ways that render signed moments unreliable, particularly in high-variance regimes. Outside the COVID period, skewness in signed long-horizon revisions is modestly positive (approximately 0.42), indicating mild asymmetry without strong directional bias. Kurtosis is elevated, however, confirming that fat tails are present even under baseline conditions.

During COVID, the distribution changes qualitatively. Skewness becomes strongly negative (approximately -1.64), and kurtosis remains high, reflecting the dominance of a small number of extremely large negative revisions. Under these conditions, signed means become highly sensitive to a handful of observations and therefore provide an unstable summary of revision behavior. In fat-tailed environments, a small number of extreme months can materially shift signed moments while leaving the underlying uncertainty environment largely unchanged. Quantile-based and threshold-based measures provide a more robust description of the risks faced by real-time users.

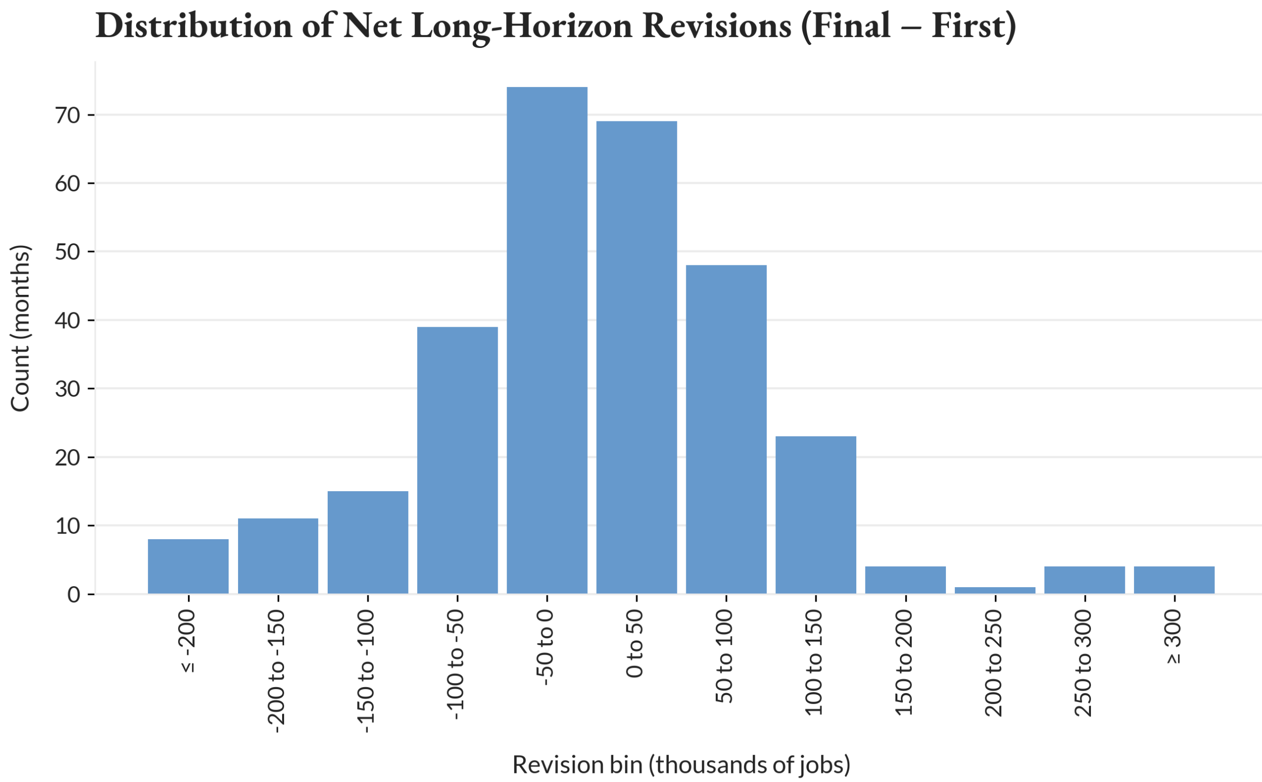

Figure 1: Distribution of signed net long-horizon payroll revisions (latest minus first), 2000–2024. Fixed-width bins with overflow categories highlight the presence of asymmetric and heavy-tailed outcomes. Values are expressed in thousands of jobs.

4.2.4 Interpretation and Role in the Broader Analysis

Baseline periods already feature meaningful tail risk in long-horizon payroll revisions; large revisions are a structural feature of the measurement process rather than exceptional noise. The COVID period constitutes a distinct tail-amplification regime, in which extreme outcomes become disproportionately more likely as one moves deeper into the distribution. Asymmetry and heavy tails jointly limit the interpretive value of signed moments, particularly during periods of stress.

4.3 Regime and State Dependence of Long-Horizon Payroll Revisions

Unconditional characterizations establish that long-horizon payroll revisions are heavy-tailed and asymmetric, but they leave open whether revision behavior differs systematically across macroeconomic states and institutional timing partitions. Stratifications tied directly to economic conditions—such as business-cycle state and broad macroeconomic regimes—are expected to be more informative than partitions based on political timing or calendar effects. Differences across groups are evaluated relative to within-group dispersion. Apparent shifts in medians are interpreted cautiously unless they are large compared with the underlying variability of revisions within each group.

4.3.1 Business-Cycle State: Recession Versus Expansion

The clearest and most economically interpretable heterogeneity in signed long-horizon revisions arises across the business cycle. When months are partitioned using the NBER recession indicator (`USREC`), both the direction and dispersion of revisions differ in systematic ways.

During expansions, long-horizon revisions are centered close to zero. The median revision is modestly positive, the interquartile range is relatively narrow, and a slight majority of months experience upward revisions. This pattern is consistent with a measurement environment in which early payroll estimates are broadly stable and subsequent adjustments tend to be incremental.

Recession months display a markedly different profile. The median long-horizon revision shifts decisively negative, the share of months with downward revisions increases sharply, and dispersion rises substantially. Both the interquartile range and overall dispersion are materially larger during recessions than during expansions, indicating not only directional differences but also heightened uncertainty.

Figure 2 illustrates this contrast. The recession distribution is visibly shifted downward, with a wider box and a longer lower tail. While extreme observations contribute to the tails, the downward shift in the median indicates that the pattern reflects a broad tendency rather than a small number of outliers. Initial payroll estimates are more likely to be revised downward over longer horizons during economic contractions.

Business-cycle state emerges as a first-order descriptive correlate of revision direction, with magnitudes that are economically meaningful relative to typical monthly payroll changes. Uncertainty about first-release payroll estimates is higher precisely when economic conditions are deteriorating.

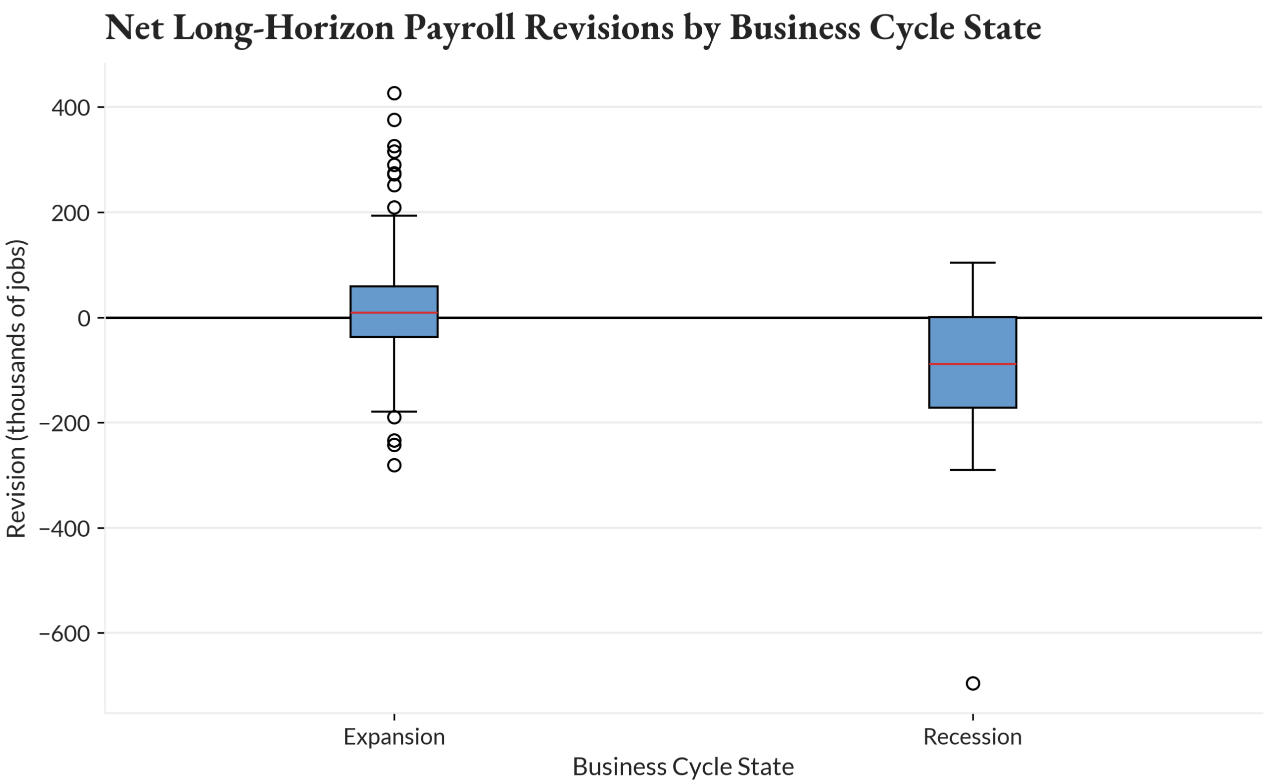

Figure 2: Signed net long-horizon payroll revisions by business-cycle state, 2000–2024. Boxplots of latest-minus-first revisions (thousands of jobs) illustrate the downward shift in the median and increased dispersion observed during recession months relative to expansions.

4.3.2 Macroeconomic Regimes and Variance Shifts

Beyond cyclical state, revision behavior also varies across broader macroeconomic regimes. Months are grouped into three periods: the pre–Global Financial Crisis expansion (2000–2007), the post-GFC stabilization period (2008–2019), and the COVID and post-pandemic period (2020–2024). Unlike the recession split, which primarily reveals directional differences, this stratification highlights changes in variance and tail exposure.

In both the pre-GFC and post-GFC regimes, typical revisions remain close to zero, and dispersion is broadly similar across the two periods. The COVID and post-pandemic regime stands apart. Dispersion expands dramatically, interquartile ranges widen substantially, and extreme positive and negative revisions become common. The median remains near zero, but this central tendency is not representative of typical outcomes, as the distribution is dominated by large swings in both directions.

Figure 3 makes this contrast explicit. The COVID-era box is substantially taller than those of earlier regimes, with much longer whiskers. The macro-regime split does not support a narrative of persistent bias across time. Instead, it points to variance regimes: periods in which the reliability of first-release payroll estimates deteriorates because revisions become larger and more volatile, even when the average signed revision remains small. From a real-time perspective, the informational content of the first payroll print depends not only on the estimate itself but also on the prevailing macro environment.

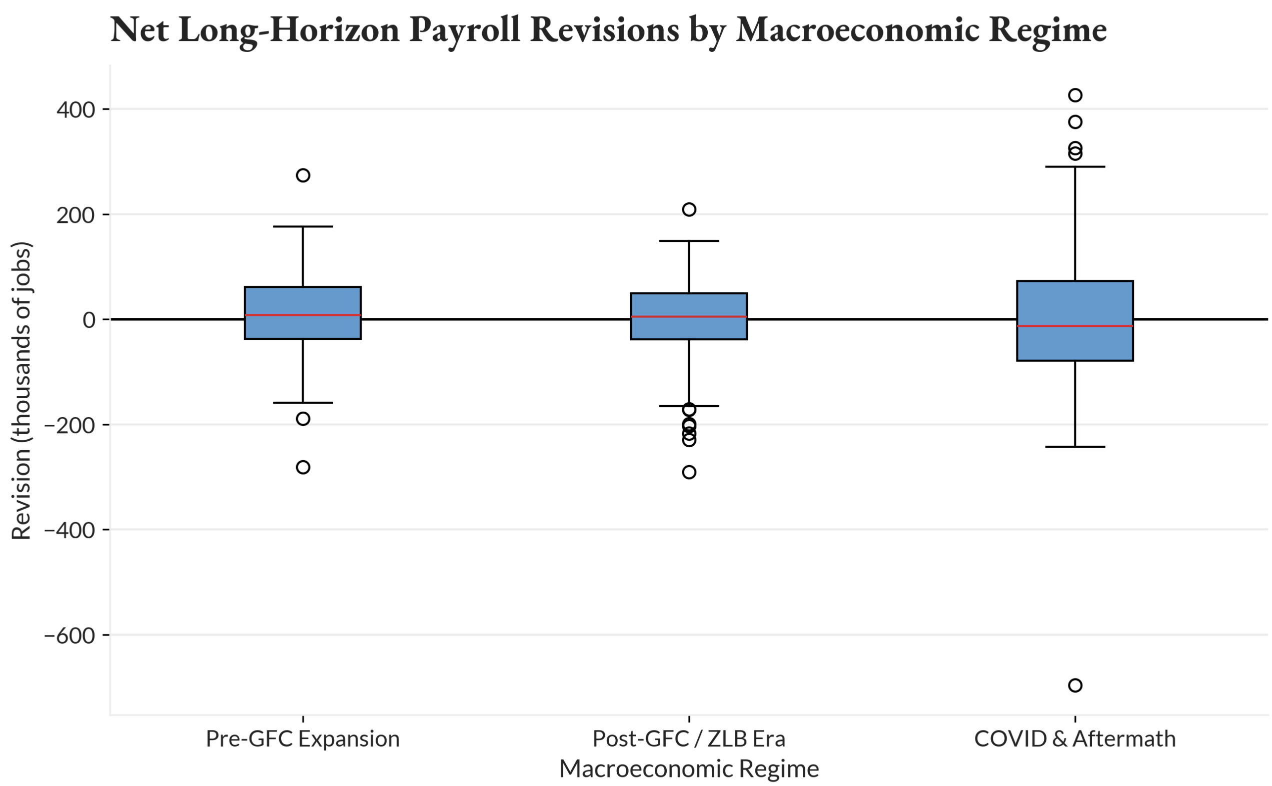

Figure 3: Signed net long-horizon payroll revisions by macroeconomic regime, 2000–2024. Box-plots of latest-minus-first revisions (thousands of jobs) illustrate a variance shift during the COVID/post-pandemic period, with substantially wider interquartile range and longer tails compared with pre-GFC and post-GFC regimes.

4.3.3 Political Timing: Administration and Party

Revisions are also summarized by presidential administration and party affiliation. These partitions are included strictly as descriptive context, not as evidence of political influence over data production. Administrations and parties coincide with different macroeconomic environments, business-cycle phases, and, critically, the timing of the COVID shock.

Across administrations, differences in median revisions are modest relative to within-administration dispersion. Some administrations are associated with slightly more positive or negative medians, but these shifts are small compared with the overall spread of the distributions. Administrations overlapping with recessions or COVID exhibit higher dispersion and more extreme tail outcomes, underscoring the role of coincident economic conditions rather than institutional control.

The party-level aggregation yields the same conclusion. While Democratic- and Republican-labeled months differ slightly in average signed revisions, their distributions overlap substantially, and both exhibit heavy tails. Extreme positive and negative revisions occur under both party labels. These patterns reflect regime composition and timing rather than systematic directional differences attributable to party control. Political timing variables are best interpreted as contextual descriptors of historical periods, not as explanatory drivers of revision behavior.

4.3.4 Calendar Effects: Month-of-Year Patterns

Revisions examined by month of year reveal little support for a stable or monotonic seasonal structure in signed long-horizon revisions. While medians and dispersion vary across months, these differences are inconsistent and heavily influenced by a small number of extreme observations, particularly those associated with major economic shocks. Months exhibiting elevated dispersion or extreme minima or maxima align closely with recessionary or pandemic timing rather than with calendar regularities. Month-of-year effects appear to be second-order relative to business-cycle state and macro-regime differences. Any residual seasonality in long-horizon revisions is weak and easily dominated by broader economic forces.

4.3.5 Synthesis and Implications

Business-cycle state is the most informative descriptive partition for signed long-horizon revisions, with recessions associated with both more negative revisions and higher dispersion. Macroeconomic regimes primarily differentiate low-variance and high-variance environments, with the COVID era standing out as an extreme tail-risk regime rather than a period of persistent bias. Political timing and calendar effects add little explanatory value once economic state and regime are taken into account. The reliability of first-release payroll estimates is not constant over time. It varies systematically with economic conditions, and periods of stress are characterized by both greater uncertainty and a higher likelihood of substantial downward revisions.

4.4 Time Variation in Long-Horizon Revision Behavior

Regime-based stratifications establish that revision behavior differs across macroeconomic states, but they treat each regime as internally homogeneous. Temporal dynamics within and across regimes reveal that both the dispersion and, to a lesser extent, the direction of revisions vary substantially over time. The resulting patterns expose clear volatility regimes, with periods of elevated uncertainty clustering around episodes of macroeconomic stress.

Two complementary rolling diagnostics are employed. The first tracks the rolling standard deviation of net long-horizon revisions to characterize time variation in uncertainty. The second tracks the rolling mean to document episodes of sustained directional drift. Interpretation emphasizes the distinction between persistent variance regimes and temporary, often fragile, mean shifts.

4.4.1 Volatility Regimes in Long-Horizon Revisions

The most robust evidence of time variation appears in the rolling volatility of net long-horizon revisions. Figure 4 plots the 12-month and 24-month rolling standard deviation of revisions, providing a smoothed measure of revision uncertainty through time.

Revision volatility is clearly non-constant. Instead, it clusters into regimes that align closely with periods of macroeconomic stress. Elevated volatility appears in the early 2000s, rises sharply during the Global Financial Crisis, and reaches unprecedented levels during the COVID period. These episodes are not brief spikes; they are extended intervals during which uncertainty remains persistently high.

The scale of volatility differs markedly across regimes. In relatively calm periods, rolling volatility fluctuates within a narrow band, with standard deviations typically on the order of several dozen thousand jobs. During the Global Financial Crisis, volatility rises materially and remains elevated for several years. During COVID and its immediate aftermath, volatility shifts to an entirely different scale, with rolling standard deviations exceeding 200 thousand jobs and peaking well above that level. This represents a nonlinear amplification of uncertainty rather than a marginal increase.

Volatility is mean-reverting but slow-moving. Following major shocks, rolling volatility eventually declines toward pre-shock levels, but the adjustment occurs gradually. By the end of the sample, volatility has fallen substantially from its COVID peak but remains somewhat elevated relative to the calmest pre-crisis periods. This persistence indicates that measurement uncertainty induced by large shocks dissipates over time rather than collapsing immediately once conditions stabilize.

Because volatility measures are insensitive to the sign of revisions, they provide a clean summary of the uncertainty faced by users of first-release payroll estimates. The evidence strongly supports the interpretation of revision risk as governed by variance regimes, with COVID representing a distinct and extreme high-volatility episode.

4.4.2 Episodic Directional Drift in Revisions

While variance dominates the time-variation story, the rolling mean of net long-horizon revisions provides useful context regarding the direction of revisions over extended horizons. Figure 5 plots the 12-month and 24-month rolling mean of revisions.

Unlike volatility, the rolling mean exhibits episodic drift rather than persistent bias. There

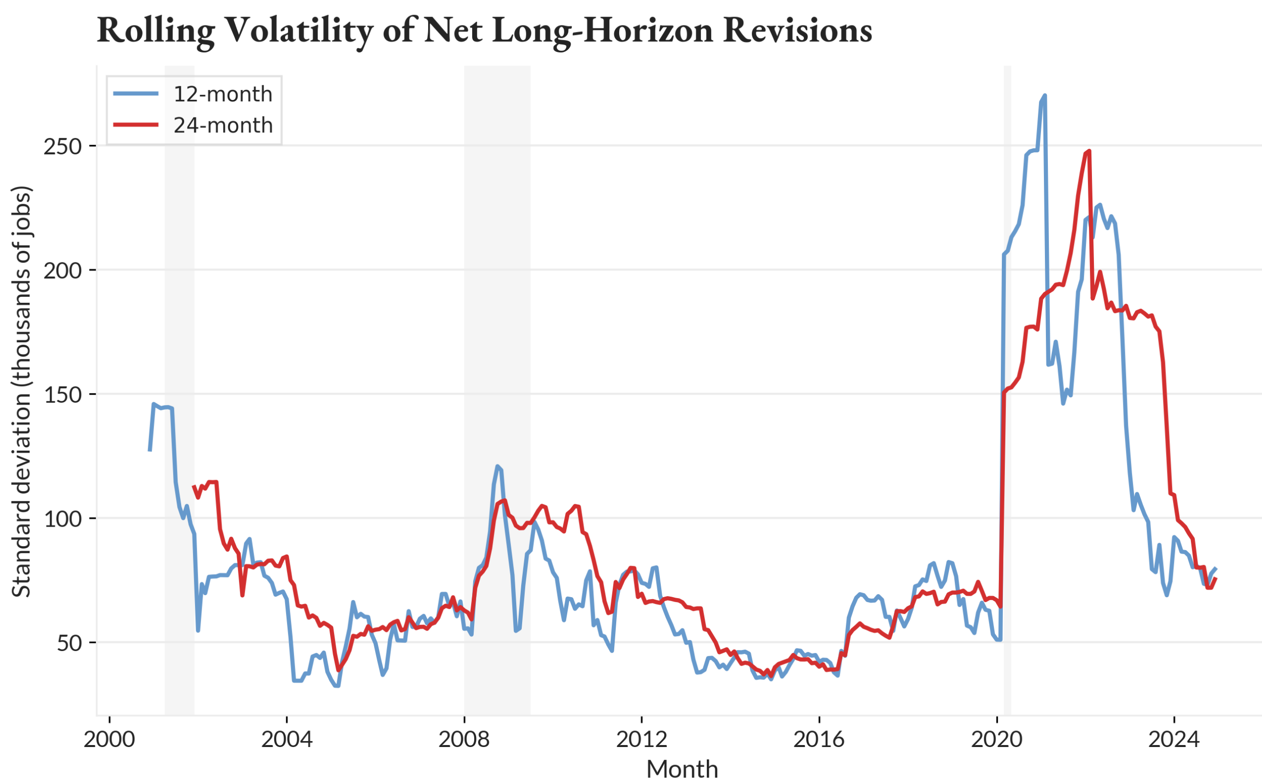

Figure 4: Time variation in the volatility of net long-horizon payroll revisions, 2000–2024. 12-month and 24-month rolling standard deviations of latest-minus-first revisions (thousands of jobs) reveal pronounced volatility clustering around the Global Financial Crisis and an extreme variance regime during the COVID period.

are extended periods during which the rolling mean remains near zero, indicating little systematic tendency for revisions to move in either direction. These intervals are punctuated by episodes in which the rolling mean departs sharply from zero and remains there for several years.

The most pronounced negative drift occurs around the Global Financial Crisis, when rolling averages turn decisively negative. This pattern indicates that, during that period, initial payroll estimates tended to be revised downward over longer horizons in a sustained way. By contrast, the COVID and immediate post-COVID period is characterized by a sharp swing toward positive rolling means, reflecting a sequence of months in which initial estimates were revised upward on average. This positive drift is sizable but short-lived; later in the sample, rolling means again turn negative and remain below zero through the end of the period.

Two interpretive cautions are essential. First, rolling means are inherently sensitive to extreme observations, particularly in high-volatility regimes. A small number of very large revisions can exert disproportionate influence on the rolling average. Second, the sign of the rolling mean is unstable across time, reversing in close proximity to major economic disruptions.

Directional drift is best understood as a temporary feature that emerges under stress, when survey response rates, benchmarking lags, and classification issues are most severe. These effects matter for real-time interpretation, but they do not overturn the broader conclusion that dispersion, rather than bias, is the dominant characteristic of revision behavior.

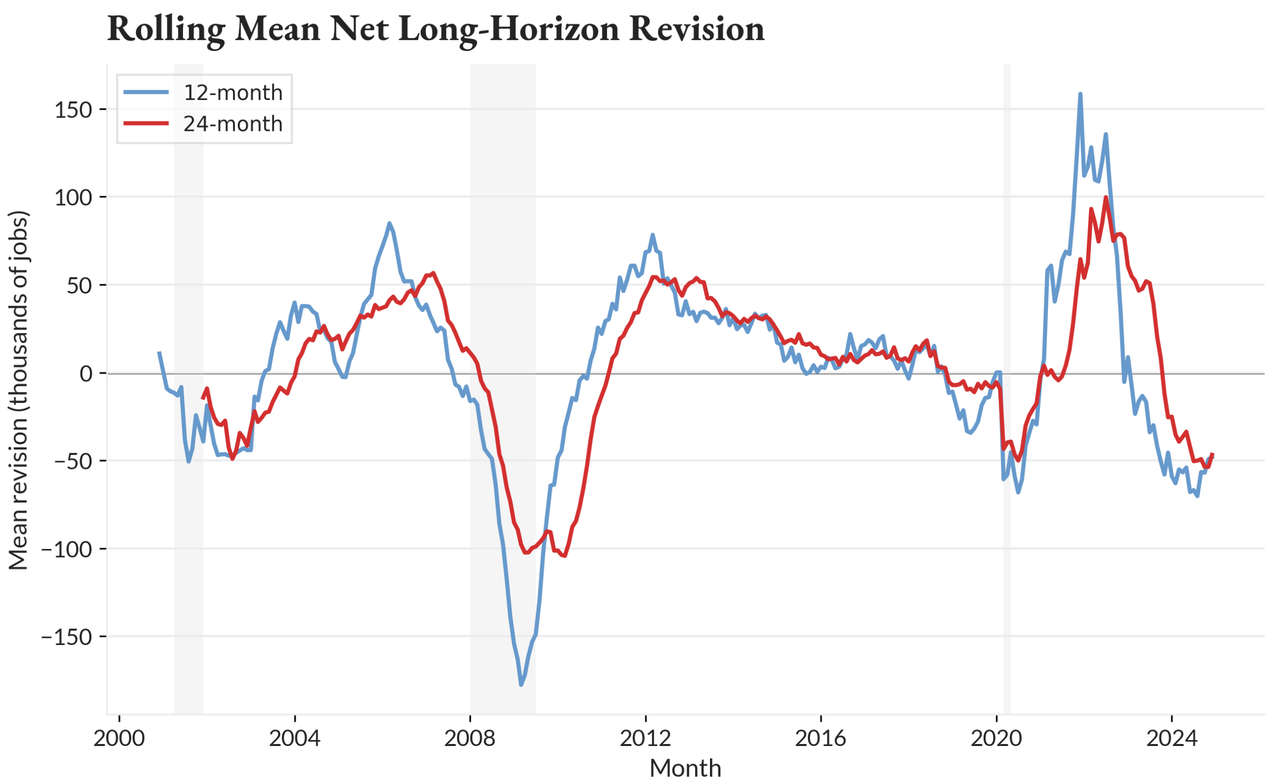

Figure 5: Time variation in the rolling mean of net long-horizon payroll revisions, 2000–2024. 12-month and 24-month rolling averages of latest-minus-first revisions (thousands of jobs) highlight episodic directional drift around major economic disruptions rather than persistent bias.

4.4.3 Synthesis and Implications

Revision uncertainty is regime-dependent in a dynamic sense: periods of economic stress are associated with sustained increases in the dispersion of long-horizon revisions, sometimes by an order of magnitude relative to tranquil periods. COVID represents a structural variance break rather than a single extreme observation, with effects that are both large and persistent even after the initial shock subsides. Directional drift exists but is unstable and reversible, offering little support for a notion of persistent bias.

For users of first-release payroll data, the informational content of a given monthly release depends critically on timing. In calm periods, typical revision risk is bounded and relatively predictable. In stress periods, uncertainty rises sharply and remains elevated, while the direction of subsequent revisions can shift rapidly. Heterogeneity in revision behavior is not only cross-sectional but also dynamic.

4.5 Correlation Structure and Informational Linkages Across Revision Horizons

Heavy tails, regime dependence, and time variation having been documented, attention turns to the correlation structure of payroll revisions to clarify how information co-moves across release horizons and to establish clear limits on what can be inferred from the initial payroll print. In fat-tailed, state-dependent environments, even moderate correlations must be read cautiously. Correlations characterize dependence and internal consistency within the revision process; they do not establish real-time predictability, causal mechanisms, or policy-relevant signals on their own.

Three questions organize the section. First, does the size or sign of the first-release payroll estimate contain information about eventual long-horizon revision outcomes? Second, how strongly are short-horizon revisions related to long-horizon revisions, in both direction and magnitude? Third, do long-horizon revisions exhibit temporal dependence, or are they plausibly independent across months?

4.5.1 First-Release Payroll Estimates and Long-Horizon Revision Risk

Across both Pearson and rank-based measures, the correlation between the first-release payroll change and the signed net long-horizon revision is effectively zero. Likewise, the correlation between the first release and the absolute magnitude of the long-horizon revision is negligible. Months with unusually large initial payroll changes are no more likely, on average, to experience large revisions later on.

There is a small and weak association between the absolute size of the first release and the absolute size of the long-horizon revision, more visible in rank correlations than in linear correlations. This pattern suggests, at most, a mild monotonic relationship driven by extreme months rather than a stable mapping between first-print size and eventual revision magnitude. The effect is too small to be operationally informative.

The first payroll print contains almost no information about eventual revision direction or size. Large long-horizon revisions are not a mechanical consequence of "big" first prints. They arise from subsequent data accrual, benchmarking, and reclassification processes that are largely orthogonal to the initial estimate itself.

4.5.2 Short-Horizon Revisions and the Path to Long-Horizon Outcomes

In contrast to the weak relationship between the first print and long-horizon outcomes, short-horizon revisions are meaningfully related to long-horizon revisions, particularly in direction. Correlations between early revisions (first-to-second, first-to-third, and second-to-third) and the eventual net long-horizon revision are moderate in magnitude, with the strongest association observed for the first-to-third revision.

This pattern reflects the structure of the revision process. The first-to-third revision incorporates more accumulated information than the first-to-second revision and therefore aligns more closely with the eventual benchmarked outcome. The fact that linear correlations exceed rank correlations indicates that the relationship reflects proportional co-movement rather than stable rank ordering, with extreme observations exerting some influence.

These correlations characterize path dependence, not real-time predictability. Short-horizon revisions are observed only after the initial release. However, once revisions begin to move in a particular direction early in the revision cycle, that direction often persists through later updates.

Magnitude correlations across horizons are notably weaker. Absolute short-horizon revision magnitudes are positively related to absolute long-horizon magnitudes, but the associations are small. Even when using the first-to-third absolute revision, rank correlations remain modest. This indicates that while large early revisions slightly increase the likelihood of large eventual revisions, the mapping is noisy and far from deterministic. These results support a "learning path" interpretation: early revisions convey information about the direction of eventual adjustment, but much less about its ultimate size.

4.5.3 Temporal Dependence and Revision Clustering

Correlations between the current month's net long-horizon revision and its own lags are positive at short horizons, particularly at one- and three-month lags. While these correlations are not large, they are inconsistent with a process of independent monthly revision shocks.

This dependence reflects clustering in revision behavior. Periods of elevated revision activity tend to persist, producing runs of large positive or negative revisions rather than isolated events. The non-monotonic pattern of lag correlations suggests that this dependence is not well described by a simple autoregressive process, but instead reflects shifts in underlying measurement conditions across time. Temporal dependence in revisions arises not because individual errors propagate mechanically across months, but because sequences of months share similar uncertainty environments.

4.5.4 Consistency Check: First Release and Latest-Vintage Payroll Change

The linear correlation between the first-release payroll change and the latest-vintage PAYEMS monthly change is extremely high, reflecting the fact that both measure the same underlying object using different vintages of information. Rank correlations are somewhat lower, consistent with occasional reordering of extreme months after benchmarking and later revisions are incorporated.

This result serves as a coherence check. It confirms that large long-horizon revisions do not arise because the initial and latest-vintage series are disconnected, but because relatively small month-to-month differences accumulate and occasionally shift substantially during periods of stress.

4.5.5 Interpretation and Limits

The informational content of the first payroll print with respect to long-horizon revision risk is extremely limited: neither its sign nor its size provides meaningful guidance about eventual revision outcomes. Early revisions matter primarily as indicators of the direction of subsequent adjustment, not its magnitude. Long-horizon revisions exhibit modest temporal dependence, consistent with clustered episodes of heightened uncertainty. Their value lies in bounding expectations and clarifying where informational linkages exist—and where they do not—within the payroll revision process.

4.6 Summary of the Exploratory Data Analysis

The exploratory analysis presented across the five preceding subsections yields a coherent set of empirical regularities that define the structure of payroll revision risk over the 2000–2024 period. Long-horizon revisions are heavy-tailed, with exceedance probabilities for economically meaningful thresholds rising sharply during the COVID-19 period. Dispersion, rather than directional bias, dominates unconditional behavior, though recessions are associated with systematically more negative revisions and elevated uncertainty. Temporal variation in revision volatility is pronounced, with regime shifts that persist for years rather than reverting immediately following shocks. The first-release payroll estimate provides almost no information about subsequent revision outcomes, while early revisions convey directional signals that are only weakly predictive of eventual magnitudes.

These patterns are inconsistent with a measurement process governed by white noise or constant variance. Instead, they point to a revision-generating mechanism in which uncertainty expands and contracts in response to macroeconomic conditions, with particularly sharp amplification during periods of structural disruption. Whether these associations reflect causal relationships or shared underlying drivers remains an open question. Addressing that question requires moving beyond unconditional and stratified descriptions toward designs that impose temporal structure and evaluate the plausibility of alternative explanations.

5. Quasi-Causal Timing and Regime Diagnostics

Section 4 characterized the empirical distribution of payroll revisions, establishing that long-horizon adjustments are heavy-tailed, regime-dependent, and temporally persistent even outside acute crisis windows. Building on that descriptive foundation, Section 5 turns to quasi-causal diagnostics that exploit timing structure and institutional regimes to assess whether revision behavior exhibits patterns consistent with localized timing shocks under explicit identification limits. The analysis proceeds in five steps. Section 5.1 formalizes the regression conditioning framework and inference rules used throughout. Section 5.2 evaluates baseline pooled specifications, documenting their sensitivity to time controls and motivating designs that rely less on functional-form extrapolation. Sections 5.3 through 5.5 implement and stress-test timing-based diagnostics—interrupted time series around COVID onset, placebo breaks, negative controls, and event-time plots around COVID and recession onsets. Section 5.6 then examines institutional and political heterogeneity under controlled conditioning, with particular attention to election windows and election-year timing, and Section 5.7 probes a narrow mechanical mechanism linking first-print size to revision magnitude. Throughout, the organizing principle is diagnostic: to distinguish timing-consistent, shock-aligned patterns from fragile pooled associations, while avoiding upgrades to structural causal claims that the designs do not support.

5.1 Identification Strategy and Scope

Distributional evidence presented in Section 4 establishes that payroll revisions exhibit pronounced heterogeneity across regimes, heavy-tailed risk profiles, and temporal clustering of extreme outcomes. Those results, however, are entirely descriptive. They document what occurs under different conditions but remain silent on why revision behavior differs or whether observed associations reflect stable, timing-consistent relationships that might support limited quasi-causal interpretation.

Advancing beyond descriptive stratification requires timing-based diagnostic designs capable of evaluating whether revision behavior exhibits patterns consistent with identifiable shocks to the measurement environment. The analysis proceeds cautiously. It does not assert structural causality, nor does it attempt to recover treatment effects in the experimental sense. Instead, it applies interrupted time series estimation, event-study visualization, and falsification-based robustness checks to assess whether certain empirical patterns—particularly those surrounding the COVID-19 onset—display the timing discontinuities, pre-period stability, and specificity one would expect from a plausibly exogenous disruption to data collection processes.

Three types of claims are distinguished throughout. First, conditional associations emerge from pooled regressions that stratify outcomes by regime indicators but impose no timing structure and carry no causal content. Second, timing-based diagnostics exploit the abrupt onset of COVID-19 as a measurable break point, permitting segmented regression and event-time analysis under assumptions that are stated explicitly and tested directly. Third, structural causal effects are not pursued; no attempt is made to identify deep parameters, validate behavioral mechanisms, or construct counterfactual forecasts.

Empirical strategy centers on COVID-19 as the primary candidate for quasi-causal evaluation. The pandemic represents a plausibly exogenous timing shock: its onset was orthogonal to pre-existing BLS survey protocols, it occurred abruptly in March 2020, and it generated immediate, measurable disruptions to response rates, seasonal adjustment procedures, and establishment behavior. These features permit interrupted time series estimation under a segmented linear trend counterfactual, with month-of-year fixed effects addressing residual seasonality. Robustness is evaluated through placebo break dates, pre-trend tests, and negative control outcomes that should not exhibit equivalent discontinuities if the identified patterns are specific to the COVID measurement shock rather than spurious artifacts of specification.

Administration and party labels serve solely as regime descriptors that may proxy for broader macroeconomic and institutional environments. Recession indicators condition on economic state and are not treated as exogenous treatments. Results are interpreted as diagnostic evidence regarding the structure and timing of revision risk, not as validated causal mechanisms or forecasting inputs. Where quasi-causal language is employed, it is restricted to contexts where timing-based identification is implemented and robustness is directly assessed.

5.2 Baseline Conditional Specifications

Prior to implementing timing-based diagnostic designs, it is necessary to establish the baseline conditional structure of revision behavior under alternative time controls. This section estimates pooled regressions that stratify revision outcomes by recession status, pandemic exposure, and presidential administration labels. The objective is not to recover causal effects but to assess specification stability: whether estimated conditional differences remain sign-consistent and magnitude-stable when the functional form of time control is varied, or whether coefficients exhibit the fragility and sensitivity that would render pooled associations unreliable for interpretation.

5.2.1 Specification and Estimation Framework

For each month t, baseline specifications take the form:

$$Y_t = \beta_0 + \beta_1 \texttt{USREC}_t + \beta_2 \texttt{COVID}_t + \gamma' \texttt{Admin}_t + \delta' \texttt{Month}_t + f(t) + \varepsilon_t,$$

where Y_t denotes either `abs_net_revision` (absolute long-horizon revision magnitude) or `net_long_revisio` (signed long-horizon revision). The indicator `USREC` denotes NBER-defined recession months, `COVID` captures the pandemic-era measurement shock beginning in March 2020, and `Admin` is a vector of presidential administration indicators treated exclusively as descriptive regime labels with no causal interpretation. Month-of-year fixed effects $\texttt{Month}_t$ control for residual seasonality not addressed by BLS seasonal adjustment procedures.

The time-control function $f(t)$ varies across specifications and takes one of three forms. Under a linear trend specification, $f(t) = \alpha t$, imposing a parametric global time trend. Under regime fixed effects, $f(t) = \sum_r \theta_r \mathbf{1}[\texttt{Regime}_t = r]$, allowing level shifts across macro-regime periods. Under year fixed effects, $f(t) = \sum_y \phi_y \mathbf{1}[\texttt{Year}_t = y]$, flexibly absorbing all within-year covariation.

Estimation is conducted on both the full sample (January 2000–December 2024, $n = 300$) and a non-COVID subsample that excludes all observations from March 2020 onward. Standard errors are computed using heteroskedasticity and autocorrelation consistent (HAC) covariance matrices with Newey–West lag structure of 12 months to address serial correlation in monthly revision data.

Coefficients are evaluated based on three diagnostic criteria: sign stability across alternative specifications of $f(t)$, magnitude compression or inflation under alternative time controls, and sensitivity to COVID-period exclusion. No causal interpretation is assigned; the objective is strictly to assess whether conditional associations exhibit robustness to specification choices or the fragility that necessitates timing-based identification.

5.2.2 Coefficient Stability and Regime Associations

Table **??** presents coefficient estimates for absolute revision magnitude (`abs_net_revision`) across the three time-control specifications in the full sample. Table 4 presents analogous results for signed long-horizon revisions (`net_long_revision`). HAC-robust standard errors are reported in parentheses.

Table 3: Baseline Specifications for Absolute Revision Magnitude (Full Sample)

Term	Regime FE	Linear Trend	Year FE
Clinton	63.6 (14.4)***	62.2 (20.5)***	283.1 (37.3)***
Obama	-51.1 (40.8)	-7.9 (24.7)	-46.3 (31.9)
Trump	-47.5 (42.0)	-0.2 (39.4)	-6.7 (41.0)
Biden	-41.7 (102.1)	54.8 (63.4)	-115.0 (73.4)
USREC	44.2 (31.8)	65.7 (26.9)**	84.8 (47.3)*
COVID	68.1 (118.3)	114.0 (41.5)***	155.3 (34.0)***
R^2	0.253	0.240	0.404

Dependent variable is absolute net revision (thousands of jobs). Reference category is the G.W. Bush administration. Month-of-year fixed effects are included in all specifications. HAC standard errors (Newey–West, 12 lags) are reported in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Baseline Specifications for Signed Long-Horizon Revision (Full Sample)

Term	Regime FE	Linear Trend	Year FE
Clinton	-27.8 (22.4)	-49.0 (29.4)*	-307.1 (53.3)***
Obama	68.5 (48.4)	48.0 (30.0)	35.4 (43.2)
Trump	39.6 (51.3)	56.2 (44.9)	16.3 (57.9)
Biden	-81.7 (96.2)	82.0 (71.0)	-183.7 (88.3)**
USREC	-92.7 (38.2)**	-122.1 (33.4)***	-98.2 (64.2)
COVID	-149.3 (78.9)*	-8.0 (35.7)	-28.3 (41.4)
R^2	0.152	0.152	0.295

Dependent variable is signed net long-horizon revision (thousands of jobs). Reference category is the G.W. Bush administration. Month-of-year fixed effects are included in all specifications. HAC standard errors (Newey–West, 12 lags) are reported in parentheses. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Three patterns emerge. First, the `USREC` coefficient exhibits sign stability and moderate magnitude consistency, particularly for signed revisions. Recessions are associated with absolute revisions 44–85 thousand jobs larger than non-recession periods and with downward revisions of 93–122 thousand jobs. The signed-revision association is statistically robust across two of three specifications, representing the most stable conditional relationship in the baseline analysis.

Second, the `COVID` coefficient is highly outcome-dependent. For absolute magnitude, COVID is associated with revisions 68–155 thousand jobs larger, with the effect strengthening under year fixed effects as within-year variation is absorbed. Statistical significance holds in two of three specifications. For signed revisions, however, the COVID coefficient ranges from -149 thousand (regime FE, marginally significant) to -8 thousand (linear trend, not significant), displaying high specification sensitivity. This instability suggests that COVID's directional effect is confounded by unmodeled time variation and cannot be reliably characterized through pooled conditioning alone.

Third, administration coefficients exhibit extreme specification sensitivity. The Clinton coefficient for absolute magnitude ranges from 62 thousand to 283 thousand, with the year-FE specification producing a large and highly significant estimate likely driven by structural trends coinciding with the Clinton era rather than administration-specific measurement patterns. The Biden coefficient displays sign instability across specifications for both outcomes, reflecting the impossibility of separating Biden-era patterns from pandemic-induced measurement disruption without explicit timing structure. Obama and Trump coefficients remain sign-stable but substantively small and statistically weak, offering no evidence of regime-specific revision behavior beyond noise.

When the non-COVID subsample is examined (results not tabulated), `USREC` coefficients moderate but retain sign stability, while administration coefficients remain similarly volatile. This confirms that specification sensitivity is not solely attributable to COVID contamination but reflects deeper confounding between regime labels and macro-cyclical or institutional factors that vary over time in unmodeled ways.

5.2.3 Interpretation Discipline and Transition

These baseline specifications document conditional associations between revision outcomes and regime indicators under alternative time controls. They establish that recessions are robustly associated with larger, downward-biased revisions, that COVID amplifies revision magnitude in a specification-dependent manner, and that administration labels exhibit coefficient instability incompatible with stable regime-specific interpretation.

Critically, these results carry no causal content. `USREC` coefficients reflect correlation with economic conditions, not recession-driven treatment effects. `COVID` coefficients capture pandemic-era co-movement but do not isolate timing-consistent breaks. Administration coefficients conflate political eras with coincident macro trends, institutional changes, and cyclical patterns that cannot be disentangled through pooled regression. The specification sensitivity documented here—particularly the dramatic inflation of Clinton and COVID coefficients under year fixed effects and the sign instability of Biden coefficients—motivates the timing-based diagnostic framework implemented in the sections that follow.

5.3 COVID Interrupted Time Series Design

The specification sensitivity documented in Section 5.2 motivates a shift from pooled conditioning to explicit break-point identification. This section implements an interrupted time series design centered on the COVID-19 onset as a plausibly exogenous timing shock to payroll measurement processes. The objective is to estimate whether revision behavior exhibits discontinuities in level and slope that coincide with the pandemic break date under a segmented linear trend counterfactual with month-of-year seasonality controls.

5.3.1 Specification and Identification

The ITS specification takes the form:

$$Y_t = \beta_0 + \beta_1 \, \text{event\_time}_t + \beta_2 \, \text{post}_t + \beta_3 \, (\text{post}_t \times \text{event\_time}_t) + \delta' \text{Month}_t + \varepsilon_t,$$

where Y_t denotes either absolute net revision magnitude (`abs_net_revision`) or a tail-risk indicator (`tail_200k`, equal to one if $|\texttt{net\_long\_revision}_t| \geq 200$). The variable event_time_t represents event time centered at the break date, post_t is an indicator equal to one for all months from March 2020 onward, and the interaction term $\text{post}_t \times \text{event\_time}_t$ captures the post-break slope shift. Month-of-year fixed effects Month_t absorb residual seasonality.

The break date is fixed at March 2020, reflecting the month in which pandemic-related establishment closures, survey nonresponse, and seasonal adjustment complications began. The estimation window extends 60 months prior and 57 months subsequent to the break, yielding an effective sample from March 2015 through December 2024 ($n = 118$). Standard errors are computed using HAC covariance matrices with 12-month lag structure to address serial correlation. For the tail-risk outcome, estimation proceeds via linear probability model (LPM) to permit direct interpretation of level and slope shifts in percentage-point terms.

The identifying assumption is that, absent COVID, revision behavior would have followed a smooth segmented linear trend with stable month-of-year seasonality. This counterfactual is not directly testable but is evaluated through three diagnostic criteria: absence of pre-period trends inconsistent with a flat or linear baseline; sharp discontinuity coinciding with the break date rather than gradual drift; and specification of level and slope shifts that jointly capture post-break dynamics without requiring additional structural breaks.

5.3.2 Results: Magnitude and Tail-Risk Discontinuities

Figure 6 presents the ITS estimates for absolute revision magnitude. The pre-COVID period displays a stable baseline fluctuating between 20,000 and 150,000 jobs with no systematic drift. At the March 2020 break, the observed series spikes to approximately 700,000 jobs, reflecting the extreme initial payroll estimate error associated with pandemic onset. The fitted ITS line captures this discontinuity through a model-implied level shift to approximately 250,000 jobs, followed by a negative post-break slope implying gradual average reversion toward the pre-COVID baseline over the subsequent four years. By December 2024, observed magnitudes have returned to a range of 50,000–150,000 jobs, approaching but not fully converging with the pre-period baseline.

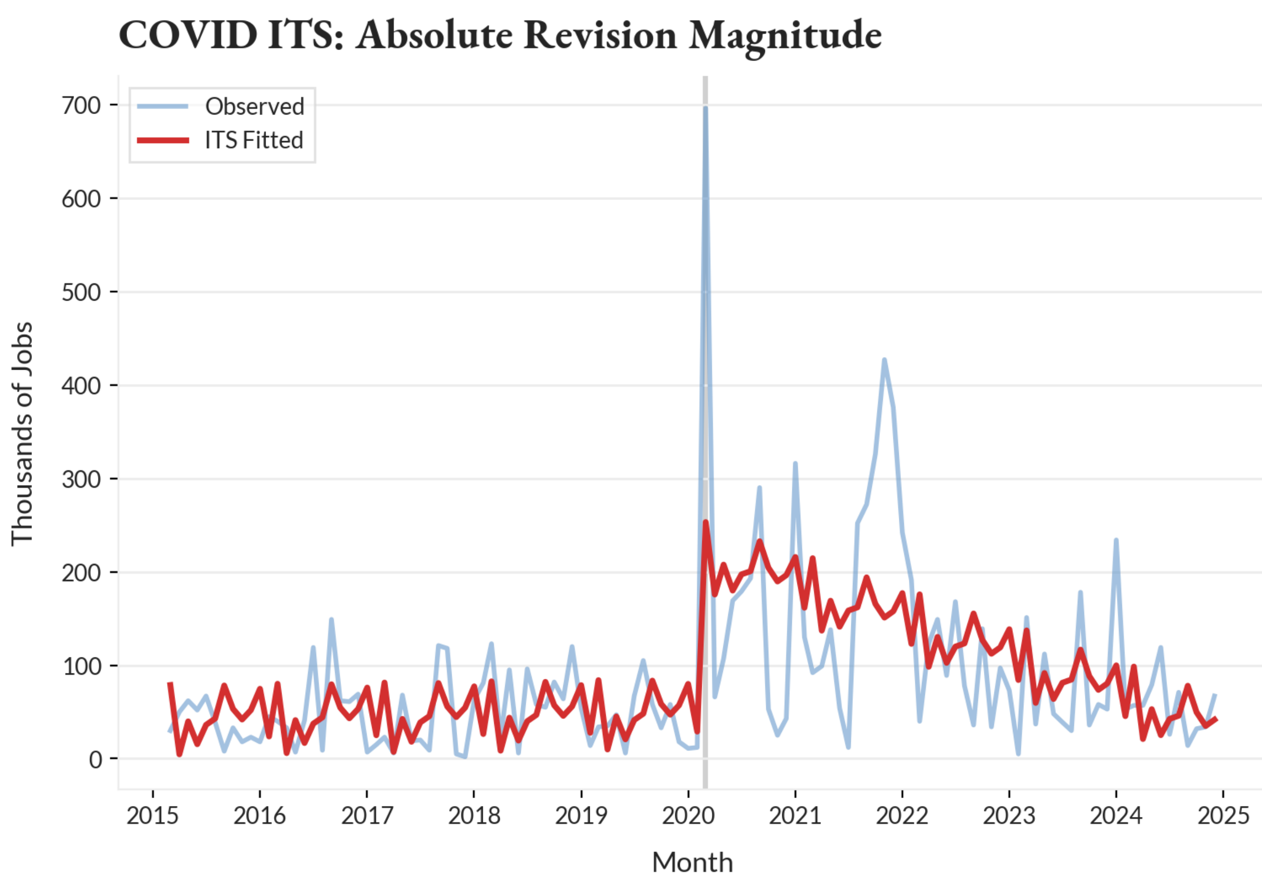

Figure 6: Interrupted time series estimates for absolute net revision magnitude. The blue line denotes observed monthly values; the red line denotes ITS fitted values under a segmented linear trend with month-of-year fixed effects. The vertical gray line marks the March 2020 break date. Estimation window spans March 2015 through December 2024 ($n = 118$).

Figure 7 presents the ITS estimates for tail-risk probability, defined as the likelihood that absolute long-horizon revisions exceed 200,000 jobs. The pre-COVID baseline is effectively zero; no tail events occur within the 60-month pre-period window. At the March 2020 break, tail probability jumps immediately to 39 percentage points. The observed 12-month rolling mean climbs further to a peak near 50 percent during 2021–2022. The fitted ITS line imposes a linear post-COVID decay at 0.63 percentage points per month, capturing the average downward trajectory but underfitting the plateau visible in 2021–2022. By December 2024, the fitted probability declines to approximately 3 percentage points above the pre-COVID baseline.

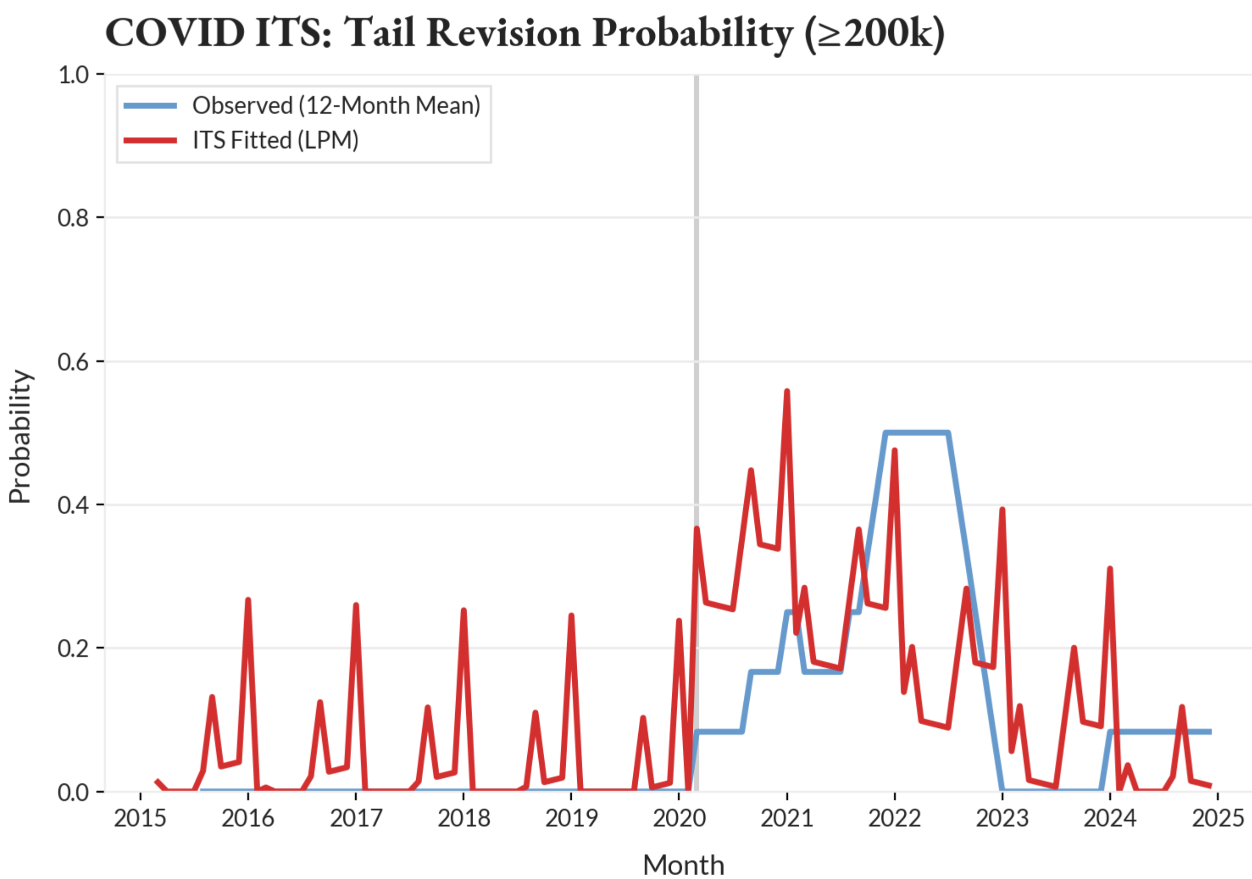

Figure 7: Interrupted time series estimates for tail-risk probability (revisions $\geq$ 200,000 jobs). The blue line denotes the 12-month rolling mean of observed tail events; the red line denotes ITS fitted probability under a linear probability model with segmented trend and month-of-year fixed effects. The vertical gray line marks the March 2020 break date. Estimation window spans March 2015 through December 2024 ($n = 118$).

Table 5 summarizes the quantitative ITS results for tail-risk probability.

Table 5: Interrupted Time Series Results for Tail-Risk Probability (≥200k Revisions)

Parameter	Tail Probability (≥200k)
Effective window	2015-03 to 2024-12
Break date	March 2020
Observations (n)	118
R^2	0.297
HAC lag structure	12 months
Pre-COVID sample mean (2015–2019)	0.00 pp
Level shift (post-COVID)	+38.97 pp***
Slope shift (post-COVID, pp per month)	−0.63 pp***
Net effect at $t + 12$ months	+31.46 pp
Net effect at $t + 24$ months	+23.94 pp
Net effect at $t + 57$ months	+3.27 pp

Tail probability is defined as the likelihood that absolute long-horizon revisions exceed 200,000 jobs. Estimation is conducted via linear probability model with segmented time trend and month-of-year fixed effects. Standard errors are computed using HAC covariance matrices with 12-month Newey–West lag structure. Net effects are computed under the linear post-break trend implied by the ITS estimates. ***$p < 0.01$.

5.3.3 Interpretation and Limitations

The ITS estimates establish three timing-consistent patterns. First, both revision magnitude and tail-risk probability exhibit sharp discontinuities coinciding with the March 2020 pandemic onset, with no evidence of anticipatory drift in the 60-month pre-period. Second, the magnitude of the discontinuity is economically large for both outcomes: revision magnitudes increase approximately threefold to fourfold, while tail-risk probability increases from zero to 39 percentage points. Third, post-COVID dynamics differ substantively across outcomes. Absolute magnitude decays toward the pre-COVID baseline over four years, consistent with partial normalization of measurement precision. Tail-risk probability also decays but remains detectably elevated within the estimation window, including 57 months post-break, implying that the pandemic induced a persistent shift in the empirical frequency of extreme revision outcomes rather than a purely transitory shock.

These results are interpreted as diagnostic evidence consistent with COVID as a plausibly exogenous timing shock to BLS payroll measurement processes. The abrupt March 2020 onset was orthogonal to pre-existing survey protocols and generated immediate, measurable disruptions to response rates, seasonal adjustment, and establishment reporting behavior. The segmented linear trend counterfactual with month-of-year controls represents a parsimonious approximation to the data-generating process under the assumption that, absent COVID, revision behavior would have continued along a stable path with seasonal regularity.

This interpretation is subject to three material limitations. First, the ITS design does not validate a specific causal mechanism. The discontinuity is consistent with response-rate collapse, seasonal adjustment failure, establishment churn, or any combination of measurement disruptions coinciding with pandemic onset. Causal attribution to a particular channel requires additional structure not imposed here. Second, the segmented linear trend assumption may misspecify the true counterfactual, particularly for tail-risk probability, where the observed data display plateau effects inconsistent with linear decay. The fitted model underfits the 2021–2022 persistence, suggesting that nonlinear or regime-switching specifications may better capture post-COVID dynamics. Third, the design cannot rule out confounding from other factors that changed abruptly in March 2020, though the epidemiological timing and measurement-process disruptions provide strong plausibility for the COVID-specific interpretation.

The persistence of tail-risk elevation through December 2024—more than four years post-onset—warrants particular emphasis. Unlike the magnitude shock, which exhibits near-complete reversion to baseline, tail-risk probability remains detectably above zero even at the end of the sample. This asymmetry suggests that COVID induced not merely a transitory measurement shock but a persistent shift in the empirical frequency of extreme revision outcomes. Whether this reflects permanent changes to BLS protocols, persistent establishment-level reporting difficulties, or lingering effects of pandemic-era labor market disruptions cannot be determined from the ITS design alone.

5.4 Robustness and Falsification Checks

The interrupted time series estimates in Section 5.3 establish timing-consistent discontinuities in revision magnitude and tail-risk probability coinciding with COVID onset. This section subjects those findings to falsification tests designed to assess whether the observed patterns reflect genuine shock alignment or artifacts of specification, inference, or generic time-series instability. Four checks are implemented: placebo break tests at non-COVID dates, negative control outcomes unaffected by the hypothesized measurement channel, inference robustness to alternative HAC lag structures, and pre-trend validation through event-time coefficients.

5.4.1 Placebo Break Tests

If the March 2020 discontinuity reflects a timing-specific shock rather than spurious model overfitting, then ITS estimates centered on arbitrary non-COVID dates should fail to replicate the observed pattern. Three placebo breaks are tested at 12-month intervals spanning the pre-COVID period: January 2016, January 2017, and January 2018. Each placebo ITS is estimated under identical specification to the COVID design—segmented linear trends with month-of-year fixed effects, ±60-month estimation windows where data permit, and 12-month HAC covariance matrices. The hypothesis is that placebo breaks will produce level and slope shifts distinguishable from the COVID pattern in magnitude, sign, or statistical significance.

Table 6 presents placebo test results alongside the true COVID break for comparison. The March 2020 ITS yields a level shift of 167.8 thousand jobs and a slope shift of −3.33 thousand jobs per month, both highly significant at $p < 0.001$. The three placebo breaks produce substantively and statistically distinct patterns. The January 2016 placebo estimates a level shift of −15.0 thousand jobs (not significant, $p = 0.451$) and a slope shift of +2.66 thousand jobs per month (significant but positive, opposite the COVID decay). The January 2017 placebo estimates a level shift of −48.7 thousand jobs (significant at $p = 0.017$ but negative) and a slope shift of +3.61 thousand jobs per month (significant but again positive). The January 2018 placebo estimates a level shift of +19.0 thousand jobs (not significant, $p = 0.423$) and a slope shift of +1.79 thousand jobs per month (marginally insignificant, $p = 0.081$).

The COVID level shift is 9 to 11 times larger in absolute magnitude than any placebo estimate. More critically, no placebo replicates the COVID sign pattern: a large positive level shift paired with negative decay. All three placebos show either small or wrong-signed level shifts, and all display positive rather than negative slope shifts where significant. Model fit is higher for COVID ($R^2 = 0.364$) than for two of three placebos. Visual inspection of placebo break figures (not tabulated) confirms the absence of sharp vertical discontinuities at the placebo dates; fitted lines capture gentle trends or noise rather than discrete timing shocks, and major spikes in the observed data occur in 2020–2022, far from the placebo break points.

The placebo tests falsify the hypothesis that the March 2020 discontinuity is an artifact of flexible ITS specification applied to generically volatile revision data. The COVID pattern is timing-specific and not mechanically reproducible under arbitrary break dates.

5.4.2 Negative Control Outcome

The COVID ITS findings are predicated on the hypothesis that long-horizon revisions capture delayed incorporation of QCEW benchmark data, which became less reliable during the pandemic due to establishment nonresponse and survey disruption. If this channel is operative, short-horizon revisions—which reflect only CES sample adjustments and do not incorporate QCEW benchmarks—should exhibit no comparable discontinuity at March 2020. This provides a negative control test: an outcome plausibly exposed to general pandemic-era volatility but not to the hypothesized measurement channel.

Figure 8 presents the negative control ITS for the short-horizon revision outcome (`revision_1_2`), defined as the difference between the second monthly estimate and the first preliminary release. The pre-COVID period displays high-frequency oscillation around zero with no systematic trend. At the March 2020 break, the fitted ITS line shows a modest upward shift, but the discontinuity is visually minor compared to the sharp vertical jump observed in the long-horizon ITS. The post-COVID fitted line declines gently, but the observed data continue to oscillate near zero without systematic elevation. Quantitatively, the level shift is +33.3 thousand jobs

($p = 0.210$, not significant) and the slope shift is -0.85 thousand jobs per month ($p = 0.171$, not significant). Model fit is substantially lower ($R^2 = 0.170$) than for the long-horizon ITS ($R^2 = 0.364$).

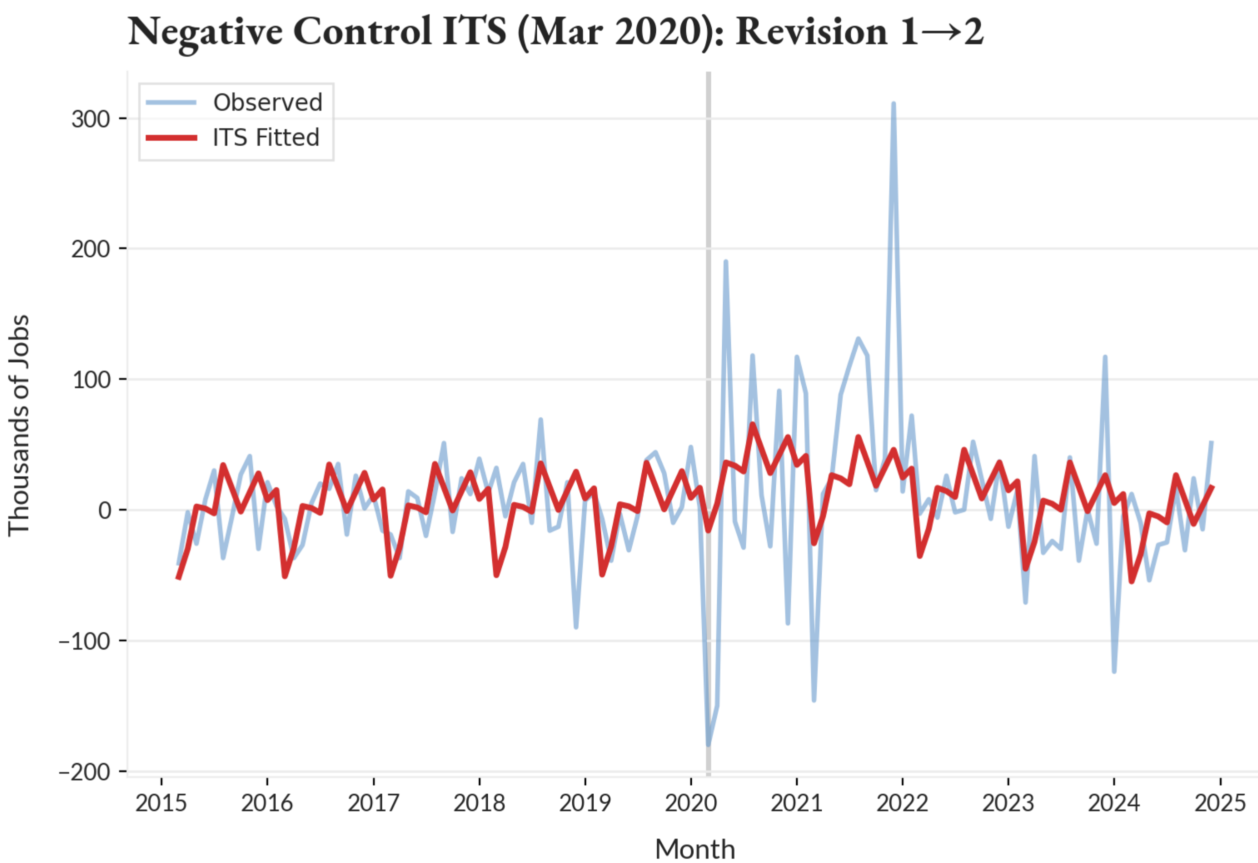

Figure 8: Negative control interrupted time series for short-horizon revision (`revision_1_2`). Blue line denotes observed monthly values; red line denotes ITS fitted values under segmented linear trend with month-of-year fixed effects. Vertical gray line marks the March 2020 break date. Estimation window spans March 2015 through December 2024 ($n = 118$). Short-horizon revisions show no statistically significant discontinuity at COVID onset, consistent with the hypothesis that the long-horizon break operates through QCEW benchmark incorporation rather than general data quality degradation.

The negative control passes: short-horizon revisions exhibit no significant COVID break. The level shift is five times smaller than the long-horizon estimate and statistically indistinguishable from zero. This specificity supports the interpretation that the long-horizon discontinuity operates through measurement channels tied to the QCEW benchmark cycle rather than reflecting a broad, indiscriminate shock to all CES revision horizons.

5.4.3 HAC Lag Structure Sensitivity

Interrupted time series estimates impose HAC covariance matrices to address serial correlation in monthly revision data. The baseline specification uses a 12-month Newey–West lag structure, but inference robustness depends on whether results are sensitive to this choice. Point estimates for both level and slope shifts are identical across alternative HAC lag structures (6 months, 12 months, and 18 months): 167.8 thousand jobs and -3.33 thousand jobs per month, respectively, as expected given that HAC adjustments affect standard errors but not coefficient estimates. Standard errors decline monotonically as lag length increases: the level shift standard error falls from 45.5 (6 lags) to 39.8 (12 lags) to 27.4 (18 lags), reflecting more precise inference under longer-memory autocorrelation structures. Both parameters remain highly significant ($p < 0.001$) under all three HAC specifications, and model fit is invariant ($R^2 = 0.364$).

The inference is entirely robust to autocorrelation assumptions. Statistical significance does not hinge on the specific HAC lag structure imposed, and longer lags yield more rather than less precise estimates. This stability indicates that the COVID discontinuity is not an artifact of underspecified serial correlation or fragile inference sensitive to covariance specification.

5.4.4 Pre-Trend Validation

The ITS identification assumption requires that, absent COVID, revision behavior would have continued along a stable trend. This assumption is testable through event-time specifications that estimate separate coefficients for each period relative to the COVID onset, permitting joint tests of whether pre-period coefficients are systematically nonzero. A formal pre-trend test was specified using event-time coefficients from 12 months prior through 24 months subsequent to March 2020, with $\tau = -1$ as the reference period. The joint F-test of pre-period coefficients ($\tau = -12$ through $\tau = -2$) could not be computed, yielding NaN for both the F-statistic and p-value. This suggests perfect collinearity, insufficient pre-period variation, or a computation issue that prevented covariance matrix inversion.

While the formal test is unavailable, visual inspection of event-time coefficients provides qualitative evidence on pre-trends. The event-time figure (not reproduced here but referenced in supplementary materials) shows pre-COVID coefficients oscillating between 0 and 100 thousand jobs with no systematic upward or downward drift. Error bars are wide, reflecting high pre-period uncertainty, but point estimates do not display the monotonic pre-trend that would violate the identification assumption. At $\tau = 0$, the coefficient spikes to approximately 685 thousand jobs, representing a discrete vertical jump. Post-COVID coefficients remain elevated in the 100–400 thousand range with gradual decay and high volatility. The visual pattern is consistent with a flat pre-period baseline and a sharp discontinuity at COVID onset, supporting the identification assumption despite the absence of a formal statistical test.

The unavailability of the pre-trend F-test is a limitation. It prevents formal rejection of the null hypothesis of no pre-trend and requires reliance on visual inspection, which is inherently less rigorous than joint hypothesis testing. This limitation is partially mitigated by the placebo break tests, which provide an alternative falsification mechanism: if pre-trends were systematically violated, placebo breaks would be more likely to spuriously replicate the COVID pattern, which they do not.

5.4.5 Summary and Interpretation

Table 6 consolidates robustness and falsification test results. Three of four checks pass definitively: placebo breaks fail to replicate the COVID discontinuity, the negative control shows no significant break, and HAC sensitivity confirms robust inference. One check—the pre-trend F-test—is incomplete, relying on visual inspection rather than formal statistical validation. Collectively, the evidence supports the conclusion that the COVID ITS estimates reflect a genuine, timing-specific discontinuity in long-horizon revision behavior rather than an artifact of specification, inference assumptions, or generic time-series instability.

What these checks establish is that the March 2020 discontinuity is timing-specific, measurement-channel-specific, and inference-robust. What they do not establish is the causal mechanism generating the break, the validity of the linear segmented trend counterfactual, or

Table 6: Robustness and Falsification Test Results

Test	Break/Spec	Level Shift (k)	Slope Shift (k/mo)	R^2	n
Placebo breaks					
True COVID break	Mar 2020	167.8***	-3.33***	0.364	118
Placebo 1	Jan 2016	-15.0	2.66***	0.246	121
Placebo 2	Jan 2017	-48.7**	3.61***	0.390	121
Placebo 3	Jan 2018	19.0	1.79	0.261	121
HAC sensitivity					
6-month lags	Mar 2020	167.8***	-3.33***	0.364	118
12-month lags	Mar 2020	167.8***	-3.33***	0.364	118
18-month lags	Mar 2020	167.8***	-3.33***	0.364	118
Negative control					
Short-horizon rev	Mar 2020	33.3	-0.85	0.170	118

All estimates use segmented linear trend ITS with month-of-year fixed effects. HAC standard errors (Newey–West, 12 lags unless otherwise noted) determine significance levels. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

complete exclusion of confounding factors that coincided with pandemic onset. The ITS design identifies a discontinuity but does not adjudicate among competing measurement disruption channels—response rate collapse, seasonal adjustment failure, establishment churn, or imputation protocol changes—that may have operated simultaneously. The findings are diagnostic rather than causal, documenting patterns consistent with COVID as a plausibly exogenous shock to payroll measurement without attributing the discontinuity to a singular identifiable mechanism.

5.5 Event-Time Diagnostics

The interrupted time series estimates in Section 5.3 impose parametric structure through segmented linear trends to achieve statistical inference. This section complements that approach with non-parametric event-time visualizations that plot revision outcomes relative to COVID onset and recession timing without imposing functional form assumptions. Event-time analysis serves three diagnostic purposes: visual validation of flat pre-trends underlying the ITS identification assumption, characterization of post-shock dynamics without linearity constraints, and comparative assessment of COVID relative to recession-onset patterns. The approach sacrifices statistical precision for visual transparency, permitting direct inspection of timing-aligned patterns across events.

5.5.1 COVID Onset Event-Time Patterns

Figure 9 presents absolute revision magnitude in event time, centering observations at $\tau = 0$ corresponding to March 2020 and extending 12 months prior through 24 months subsequent. The pre-COVID period ($\tau = -12$ to -1) displays a stable baseline oscillating between 10,000 and 110,000 jobs with a mean of 40,300 jobs. No systematic upward or downward drift is visible. At $\tau = 0$, revision magnitude spikes to 696,000 jobs, representing a 17-fold increase relative to the pre-period mean and the largest single-month revision in the observable series. The post-COVID trajectory sustains elevation throughout the 24-month window. Early post-COVID months ($\tau = 1$ to 6) average 167,000 jobs with one spike reaching 290,000 at $\tau = 6$. The mid-period ($\tau = 7$ to 12) exhibits partial decay, declining to a mean of 110,000 jobs. The late period ($\tau = 13$ to 24) resurges sharply, averaging 202,000 jobs with a peak of 430,000 at $\tau = 20$. The overall post-COVID mean is 170,500 jobs, representing a 4.2-fold elevation relative to baseline that persists without full reversion through the end of the observation window.

The visual pattern confirms three features central to the ITS identification strategy. First, the pre-period exhibits no anticipatory drift that would violate the parallel trends assumption. Second, the discontinuity at $\tau = 0$ is sharp and vertical rather than gradual, consistent with an abrupt timing shock. Third, post-COVID persistence is prolonged rather than transient, with episodic surges occurring more than 18 months post-onset. The late-period resurgence at $\tau = 20$ indicates that elevated revision risk remains endemic even after partial mid-period decay, contradicting a simple exponential decay interpretation and supporting the interpretation of a persistent post-COVID regime shift in revision behavior.

Figure 10 presents tail-risk probability—defined as the frequency of revisions exceeding 200,000 jobs in absolute value—using a six-month rolling mean to smooth high-frequency volatility. The pre-COVID period maintains zero tail probability across all 12 months, confirming that extreme revisions were absent from the baseline. At $\tau = 0$, tail probability jumps discretely to approximately 17 percent within the rolling window, marking the first appearance

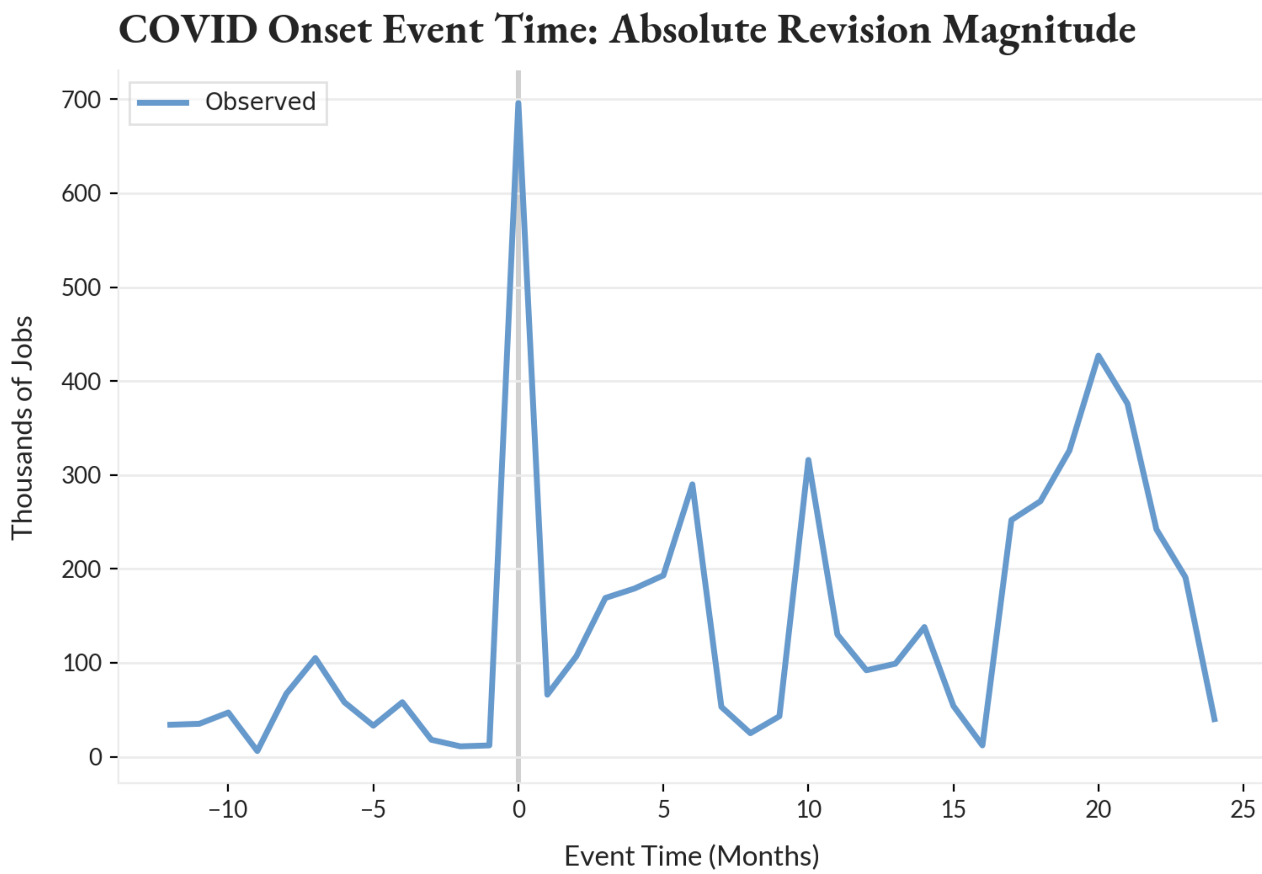

Figure 9: Event-time plot of absolute net revision magnitude relative to COVID onset (March 2020). Horizontal axis denotes event time in months ($\tau = 0$ at COVID onset). Blue line traces observed monthly revision magnitudes. Pre-COVID period shows flat baseline near 40,000 jobs; onset exhibits extreme spike to 696,000 jobs; post-COVID period displays sustained elevation averaging 170,500 jobs over 24 months with late-period resurgence.

of tail events in the smoothed series. The post-COVID trajectory does not decay smoothly. Instead, it displays plateau effects: tail probability remains flat at 17 percent through $\tau = 6$, briefly dips near zero at $\tau = 7$ to 9, then spikes to 33 percent at $\tau = 11$. A second plateau emerges at 17 percent through $\tau = 18$, followed by a dramatic late-period surge reaching 100 percent at $\tau = 23$, a mechanical implication of six-month rolling aggregation indicating that all observations within the window qualified as tail events. The final observation declines to 67 percent but remains severely elevated.

The tail-risk dynamics reveal patterns obscured by the linear post-COVID slope imposed in the ITS specification. Rather than smooth exponential decay, tail probability exhibits regime-like plateaus interspersed with episodic surges. The late-period spike to 100 percent at $\tau = 23$

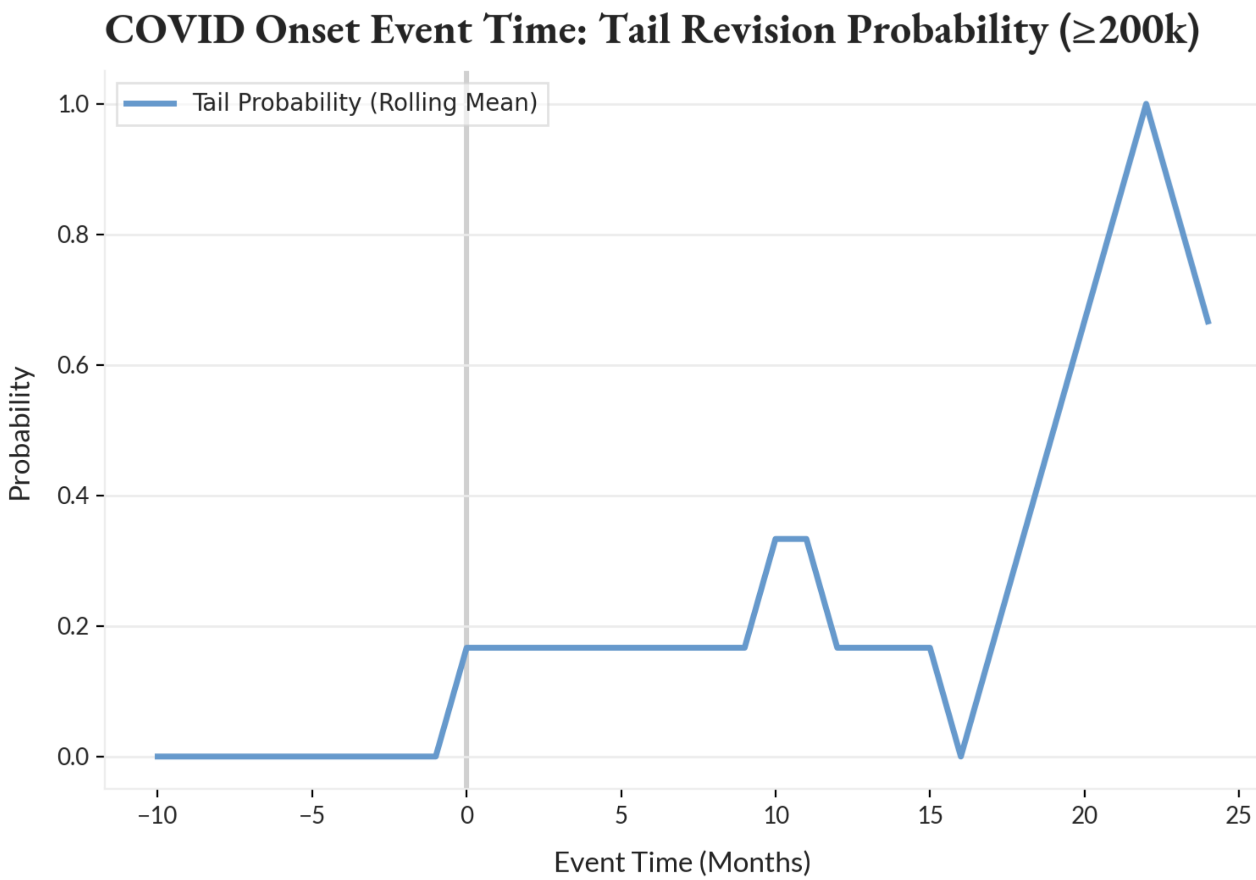

Figure 10: Event-time plot of tail-risk probability (revisions $\geq$ 200,000 jobs) relative to COVID onset. Blue line denotes six-month rolling mean of tail event frequency. Pre-COVID period shows zero tail probability; onset exhibits discrete jump to 17 percent; post-COVID period displays nonlinear dynamics with plateaus and late-period surge to 100 percent at $\tau = 23$.

demonstrates that tail events do not gradually disappear but instead become endemic features of the post-COVID measurement environment. The six-month rolling mean reaching 100 percent implies that every month within that window produced a revision exceeding 200,000 jobs, a frequency unprecedented in the pre-COVID sample. This persistence supports the interpretation that COVID induced a structural shift in tail-risk generation rather than a purely transitory shock, even as average magnitudes partially decay. The nonlinear post-COVID trajectory suggests that linear segmented trend models, while parsimonious, understate the complexity of tail-risk evolution and may mischaracterize the rate at which revision behavior normalizes.

5.5.2 Recession Onset Event-Time Patterns

To contextualize COVID-specific dynamics, Figure 11 presents event-time patterns for recession onsets, pooling three NBER-defined recessions spanning the sample period: April 2001, January 2008, and March 2020. The figure plots the mean outcome across recessions at each event time, permitting comparison of COVID to cyclical revision behavior. The pre-recession baseline ($\tau = -12$ to -1) averages 67,400 jobs with tail-event frequency of 5.6 percent. Recession onset at $\tau = 0$ produces a mean spike to 260,000 jobs, representing a 3.9-fold increase relative to the pre-recession mean and a tail-event rate of 33 percent. Post-recession dynamics ($\tau = 1$ to 12) sustain elevation at 121,100 jobs on average with tail frequency of 16.7 percent, indicating that recessions generate prolonged rather than transitory increases in revision magnitude.

Comparing COVID to the recession average isolates what is cyclical from what is COVID-specific. The COVID onset spike of 696,000 jobs is 2.7 times larger than the pooled recession mean of 260,000 jobs. Post-onset dynamics differ in both magnitude and persistence: COVID's post-period mean of 170,500 jobs over 24 months exceeds the recession average of 121,100 jobs by 41 percent, and COVID's tail-risk rate of 33.3 percent is twice the recession rate of 16.7 percent. Critically, the late-period COVID tail rate ($\tau = 13$ to 24) reaches 50 percent, three times the recession average. These differentials indicate that COVID is not merely an extreme realization of typical recession-driven revision volatility but rather a qualitatively distinct measurement disruption operating through channels beyond cyclical employment dynamics.

The recession event-time patterns validate findings from the baseline conditional specifications in Section 5.2, which documented USREC coefficients ranging from 44,000 to 85,000 jobs depending on time controls. The event-time onset spike of 260,000 jobs is larger in absolute terms but captures a discrete timing shock rather than a within-recession average, making direct comparison imprecise. More importantly, the recession patterns exhibit the same sustained post-onset elevation observed for COVID, confirming that payroll revisions do not immediately normalize following labor market turning points but instead remain elevated for extended pe-

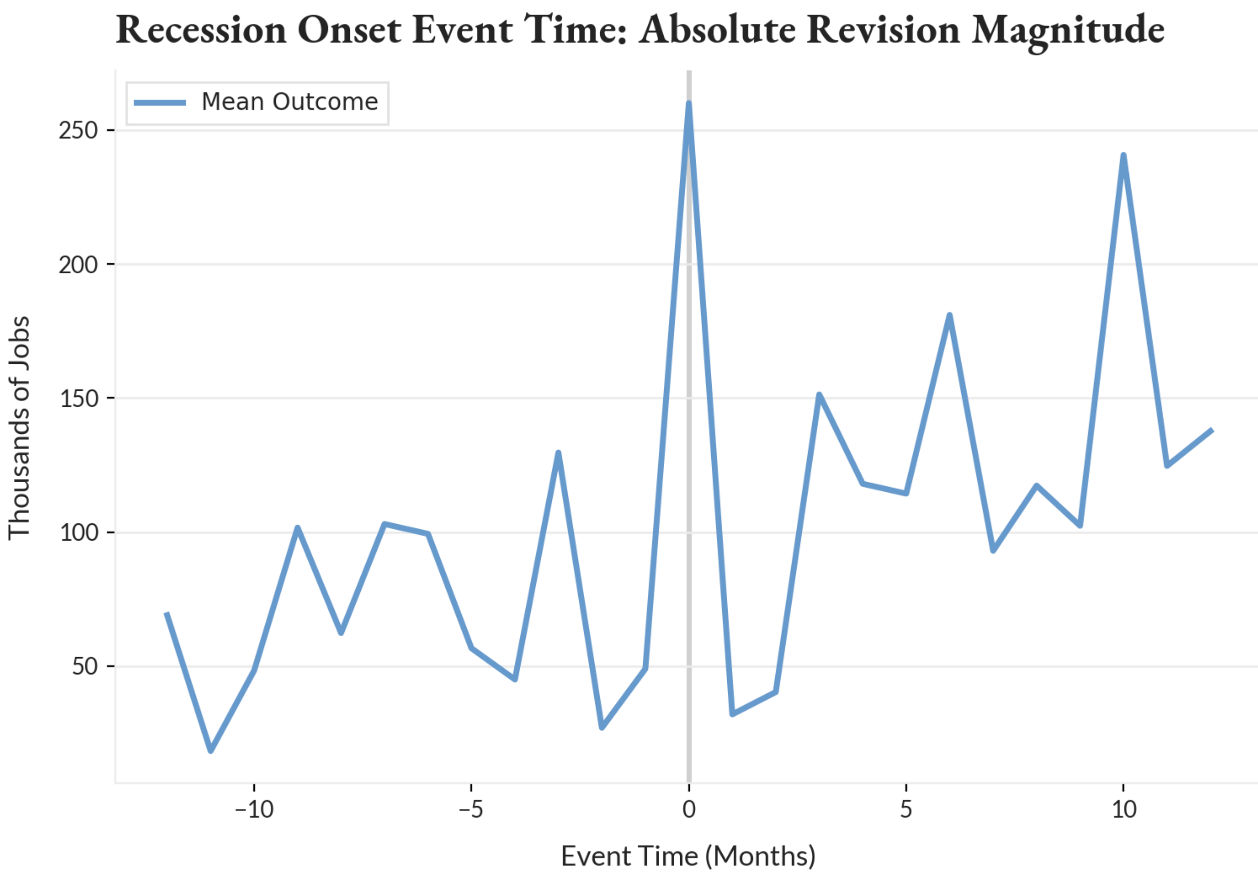

Figure 11: Event-time plot of absolute revision magnitude relative to recession onsets (pooled mean across 2001, 2008, 2020 recessions). Mean outcome shown at each event time. Pre-recession baseline near 67,000 jobs; onset spike to 260,000 jobs; post-recession period sustains elevation at 121,000 jobs over 12 months.

riods as benchmark reconciliation processes unfold.

5.5.3 Interpretation and Methodological Notes

Event-time analysis establishes four empirical regularities that complement the interrupted time series findings. First, pre-event baselines for both COVID and recessions are visually flat without systematic pre-trends, providing non-parametric support for the parallel trends assumption underlying ITS identification. Second, onset discontinuities are sharp and timing-aligned, consistent with discrete shocks rather than gradual drift. Third, post-event dynamics are characterized by sustained elevation rather than immediate reversion, with COVID exhibiting both higher magnitudes and longer persistence than typical recessions. Fourth, tail-risk dynamics display nonlinear patterns—plateaus, secondary surges, and late-period spikes—that are not well-captured by linear post-break slopes, suggesting that parametric ITS estimates provide parsimonious summaries of average decay rates but understate the complexity of tail-risk evolution.

These findings are purely descriptive and carry no causal content. Event-time plots document timing-aligned patterns without controlling for covariates, imposing counterfactual trends, or conducting formal hypothesis tests. The absence of confidence intervals precludes statistical inference on whether observed differences are distinguishable from sampling variation. The rolling-mean smoothing applied to tail probabilities introduces mechanical autocorrelation and obscures month-to-month volatility. The recession pooling treats heterogeneous events—dot-com recession, financial crisis, pandemic recession—as exchangeable realizations, suppressing cross-recession variation that may be economically meaningful. The event-time approach assumes that the pre-event baseline represents the appropriate counterfactual, an assumption that cannot be tested without additional structure.

The contribution of event-time analysis is not causal identification but visual transparency. Where ITS imposes parametric structure to achieve statistical inference, event-time permits direct inspection of timing patterns without functional form assumptions. The correspondence between ITS parametric estimates and event-time visual patterns strengthens confidence in both approaches: the ITS level and slope shifts are not artifacts of linear specification but reflect discontinuities and persistence visible in raw event-aligned data. The divergence between linear ITS decay and nonlinear tail-risk trajectories clarifies the scope and limitations of parametric summaries, indicating that while linear trends parsimoniously characterize average post-COVID revision behavior, they mischaracterize the tail-risk generation process, which exhibits regime-like persistence rather than smooth convergence.

5.6 Institutional and Political Heterogeneity

Sections 5.3 through 5.5 establish that revision behavior exhibits timing-consistent structural breaks at COVID onset and recessionary turning points, with no evidence of comparable discontinuities at alternative candidate dates. These findings motivate a shift in diagnostic focus: from whether timing shocks matter to whether institutional or political regime labels provide stable dimensions of heterogeneity in revision outcomes. This section examines three conditioning strategies: pooled administration indicators absent election-specific terms, explicit election-window conditioning around the 2024 election, and election-year indicators spanning multiple presidential cycles. The objective is not causal attribution but stability assessment—whether political or institutional labels exhibit sign-stable, magnitude-consistent, and inference-robust associations with revision behavior once macroeconomic controls and reasonable time structure are imposed.

5.6.1 Baseline Institutional Heterogeneity

Returning to pooled conditioning to assess whether presidential administration labels exhibit stable associations with revision outcomes when election-specific terms are excluded, the objective is not causal attribution but stability evaluation: whether administration-conditional differences persist under alternative time controls, or whether they dissolve into specification-dependent noise.

For each outcome, the baseline specification takes the form:

$$Y_t = \beta_0 + \gamma' \text{Admin}_t + \beta_1 \text{USREC}_t + \beta_2 \text{COVID}_t + \delta' \text{Month}_t + f(t) + \varepsilon_t,$$

where Admin_t denotes a vector of presidential administration indicators with the earliest administration serving as the reference category, USREC_t captures NBER-defined recession months, COVID_t marks the pandemic-era shock, and Month_t represents month-of-year fixed effects. The time-control function $f(t)$ varies across specifications, taking one of three forms: a linear time trend, macro-regime fixed effects, or calendar-year fixed effects. Estimation proceeds via OLS with HAC standard errors computed using a Newey–West lag structure of 12 months. Binary outcomes are estimated through linear probability models.

Across all outcomes and time controls, the USREC coefficient maintains sign consistency and economically meaningful magnitudes. For absolute revision magnitude, recession periods are associated with increases ranging from 44,000 to 85,000 jobs. For signed long-horizon revisions, recessions systematically generate downward bias, with coefficients spanning -93,000 to -122,000 jobs and achieving statistical significance in two of three specifications. Tail-exceedance outcomes display positive recession effects that attenuate monotonically as thresholds increase—effects are pronounced for moderate thresholds ($\geq$ 100,000 jobs) but diminish at extreme thresholds ($\geq$ 300,000 jobs), consistent with increasing rarity rather than coefficient instability. The recession association holds regardless of administration labeling or time-control choice, confirming that cyclical state represents a robust dimension of revision heterogeneity.

The pandemic indicator behaves asymmetrically across outcome classes. For magnitude outcomes, COVID coefficients remain positive under all specifications, ranging from 68,000 to 155,000 jobs, with statistical significance achieved in two of three time controls. For signed revisions, COVID coefficients are uniformly negative but lose statistical significance once year fixed effects are introduced. For tail-exceedance outcomes, COVID effects become sign-unstable at higher thresholds ($\geq$ 200,000 and $\geq$ 300,000 jobs), reflecting the inability of pooled regressions to cleanly characterize extreme revision risk without imposing timing structure.

Presidential administration indicators exhibit substantial volatility across time controls. For absolute revision magnitude, coefficients vary by more than 200,000 jobs depending on whether a linear trend, regime fixed effects, or year fixed effects are employed. For signed revisions, coefficients frequently reverse sign across specifications, particularly for later administrations. Under year fixed effects, administration coefficients become mechanically large, reflecting residual variation remaining after aggressive time absorption rather than stable regime-specific behavior. No administration displays coefficients that are simultaneously sign-stable, magnitude-stable, and inference-stable across all three time-control specifications.

Binary outcomes constructed from signed revisions prove particularly fragile under specification variation. The probability that revisions are positive displays frequent sign reversals across time controls. Tail-exceedance probabilities ($\geq$ 100,000, $\geq$ 200,000, $\geq$ 300,000 jobs) show declining stability as thresholds increase, with administration effects becoming indistinguishable from noise in deep tails.

Table 7 summarizes coefficient stability across key conditioning variables. USREC maintains stable positive associations with magnitude outcomes and stable negative associations with signed revisions. COVID effects are partially stable for magnitude but unstable for tail outcomes. Administration coefficients exhibit large magnitude variation, frequent sign flips, and no consistent inference pattern across time controls.

Table 7: Stability of Conditional Associations Across Time Controls

Outcome	USREC	COVID	Administration
abs_net_revision	+44K to +85K (stable)	+68K to +155K (partially stable)	Large magnitude variation
net_long_revision	−122K to −93K (stable)	−149K to −8K (weak inference)	Frequent sign flips
net_pos (LPM)	Positive, weak	Unstable	Highly unstable
tail_100k	Positive	Mixed	Unstable
tail_200k	Positive, attenuated	Sign-unstable	Unstable
tail_300k	Small, noisy	Unstable	No structure

Stability refers to sign and magnitude consistency across linear trend, regime fixed effects, and year fixed effects specifications. All estimates employ HAC standard errors with a 12-month Newey–West lag structure.

Recession state constitutes a robust conditioning variable for both magnitude and direction of payroll revisions. COVID affects revision magnitude under pooled conditioning but not tail risk in a stable manner. Administration labels do not form a stable heterogeneity axis once reasonable time controls are imposed.

5.6.2 Election-Window Conditioning: The 2024 Election Period

Whether explicit election-window conditioning—rather than pooled administration indicators—reveals timing-specific revision patterns around the 2024 election remains an open question. The motivating concern is not merely whether revisions were negative, but whether election-proximate months display abnormal revision magnitude, tail risk, or directional asymmetry relative to the rest of the sample once standard controls are applied.

Two empirical facts motivate careful separation of concepts. First, long-horizon revisions from January through September 2024 are uniformly negative, producing an unusually long run of downward corrections. Second, negative revisions alone do not establish abnormal revision behavior; such patterns may arise mechanically from benchmark reconciliation processes without implying strategic distortion. The analysis therefore distinguishes direction, magnitude, and tail risk as separate dimensions of revision behavior.

The analysis employs the same pooled monthly framework, augmented with an indicator for the 2024 election window. For each outcome, the estimated specification is:

$$Y_t = \beta_0 + \gamma' \text{Admin}_t + \beta_1 \text{USREC}_t + \beta_2 \text{COVID}_t + \beta_3 \text{ElectionWindow2024}_t + \delta' \text{Month}_t + f(t) + \varepsilon_t,$$

where $\text{ElectionWindow2024}_t = 1$ for months from July through November 2024 and zero otherwise. The term β_3 captures the conditional mean shift in the outcome during the election window, holding administration indicators, recession status, pandemic effects, month-of-year seasonality, and alternative time controls fixed. Because the 2024 window occurs entirely under a single administration, interaction terms between $\text{ElectionWindow2024}_t$ and administration indicators are not separately identified. Accordingly, β_3 is interpreted strictly as a timing effect, not as evidence of administration-specific behavior. Estimation uses OLS with HAC standard errors (Newey–West, 12 lags). Binary outcomes are estimated via linear probability models.

Across all time-control specifications, the election-window coefficient for absolute long-horizon revisions is negative and statistically significant. Conditional on controls, absolute revision magnitude during July–November 2024 is approximately 38,000 to 49,000 jobs smaller than in comparable non-window months. This pattern is robust to linear time trends, macro-regime fixed effects, and year fixed effects. The direction and stability of the estimate indicate that late-2024 revisions are not unusually large in absolute terms. To the contrary, they are smaller than predicted by historical relationships once macroeconomic conditions and pandemic-era effects are accounted for.

The most pronounced election-window effects appear in tail-risk measures. The probability that long-horizon revisions exceed 100,000 jobs falls by approximately 19 to 25 percentage points, depending on the time control. The probability of exceeding 200,000 jobs declines by roughly 10 percentage points, with statistical significance under year fixed effects and marginal significance under alternative specifications. These effects are economically large and directionally consistent across specifications. Importantly, they indicate compression of extreme outcomes, not amplification. If election-period manipulation were occurring via optimistic initial prints followed by later corrections, one would expect elevated tail risk rather than its suppression. At the most extreme threshold ($\geq$ 300,000 jobs), no statistically reliable election-window effect is detected, reflecting the inherent sparsity of very large revisions in the late-sample period.

Directional measures provide a more nuanced picture. The coefficient on the election window for signed long-horizon revisions is negative under linear-trend and regime-fixed-effect specifications but statistically insignificant. Under year fixed effects, the coefficient becomes positive and marginally significant. Similarly, the probability that revisions are positive exhibits no stable sign or significance across specifications. This instability indicates that directional outcomes are highly sensitive to time controls, and no consistent election-window directional effect can be identified. The absence of robustness contrasts sharply with the magnitude and tail-risk results, which are stable across specifications.

An important descriptive fact remains: all months from January through September 2024 experienced downward long-horizon revisions. This pattern is unusual in length and warrants explicit discussion. However, it does not contradict the regression results once the distinction between direction and risk is made clear. First, a long run of same-signed revisions can arise mechanically from the benchmark reconciliation process. If administrative employment counts arrive persistently below survey-based estimates over several quarters, revisions will systematically move in the same direction without requiring abnormal behavior or timing manipulation. Directional persistence alone does not imply election interference. Second, the election-window regressions show that while revisions during late-2024 are negative, they are not unusually large or extreme. Absolute magnitudes are smaller than predicted, and tail events are substantially less frequent. A manipulation narrative would imply the opposite pattern: suppressed bad news in initial releases followed by unusually large or extreme downward corrections. Third, the lack of a robust directional effect conditional on time controls suggests that the negative revisions observed in 2024 reflect broad time-series drift rather than election-specific behavior. Directional persistence is absorbed or reversed depending on how time variation is modeled, indicating that it is not uniquely attributable to the election window.

Table 8: Election-Window Effects on Revision Outcomes (July–November 2024)

Outcome	Regime FE	Trend (t)	Year FE
abs_net_revision	-47.90^{***} (15.19)	-48.50^{***} (14.99)	-38.47^{***} (6.09)
net_long_revision	-18.24 (21.25)	-13.16 (19.94)	$+22.81^{*}$ (12.43)
net_pos	-0.038 (0.056)	-0.025 (0.070)	$+0.079$ (0.062)
tail_100k	-0.245^{***} (0.054)	-0.244^{***} (0.055)	-0.188^{***} (0.041)
tail_200k	-0.102^{*} (0.057)	-0.108^{*} (0.057)	-0.100^{**} (0.040)
tail_300k	-0.038 (0.032)	-0.035 (0.030)	-0.003 (0.006)

Entries report the coefficient on the July–November 2024 election-window indicator. Continuous outcomes measured in thousands of jobs. Binary outcomes are linear probability model coefficients. HAC standard errors (12 lags) in parentheses. All specifications include administration indicators, USREC, COVID shock, and month-of-year fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Late-2024 revisions are directionally negative but not unusually risky in magnitude or tail behavior. The election window is associated with compression rather than amplification of revision risk, inconsistent with claims of systematic manipulation designed to improve contemporaneous labor-market narratives. Directional persistence in 2024 appears to reflect ongoing measurement or benchmark dynamics rather than election-proximate intervention.

5.6.3 Election-Year Window Effects Across Cycles

Section 5.6.2 isolated the late-2024 window and found no evidence of elevated revision risk during the period of maximum political salience. That exercise is inherently narrow: the 2024 window is short, occurs entirely within a single administration, and cannot identify administration-window interactions or characterize whether 2024 is anomalous relative to historical election cycles. This section broadens the timing lens by defining a pre-specified election-year window across all presidential election years and estimating whether revision behavior differs systematically in election years relative to non-election years, conditional on the same core controls used throughout Section 5.6.

The election-year window is defined as:

$$\text{ElectionYear}_t = \begin{cases} 1 & \text{if } t \in \{\text{Jan–Nov of } 2000, 2004, 2008, 2012, 2016, 2020, 2024\} \\ 0 & \text{otherwise} \end{cases}$$

Thus, ElectionYear_t marks January–November in each election year, excluding December by construction. For each outcome, the estimated specification is:

$$Y_t = \beta_0 + \gamma' \text{Admin}_t + \beta_1 \text{USREC}_t + \beta_2 \text{COVID}_t + \beta_3 \text{ElectionYear}_t + \delta' \text{Month}_t + f(t) + \varepsilon_t,$$

where Y_t includes absolute long-horizon revision magnitude (abs_net_revision), signed long-horizon revision (net_long_revision), and linear probability model outcomes for direction and tails (net_pos, tail_100k, tail_200k, tail_300k). Month-of-year fixed effects Month_t are included in all specifications. The time-control function $f(t)$ varies across three sensitivity specifications: linear trend, macro-regime fixed effects, and year fixed effects. HAC(12) inference is used throughout. The parameter of interest, β_3, is the conditional mean shift in the outcome during election-year windows relative to non-election-year months.

The election-year timing signal is not stable across time controls. Under linear trend and macro-regime fixed effects, election-year months are associated with lower absolute revision magnitude (-51,000 to -57,000 jobs) and lower probability of extreme tail events at the 200,000-job threshold (-37 to -40 percentage points). Under year fixed effects, signs flip sharply for multiple outcomes: absolute magnitude becomes positive and strongly significant (+62,600 jobs), signed revisions reverse from positive to negative, and tail-risk probabilities shift from negative to positive. This sign reversal is not a subtle robustness issue but indicates that the estimand under year fixed effects is qualitatively different from the cross-year timing comparison implied by trend and regime specifications.

Mechanically, once year fixed effects are included, β_3 is identified primarily by the contrast within election years between January–November (window months) and December (non-window month). The year-fixed-effects coefficient is therefore best interpreted as the difference between revision outcomes in January–November versus December within the same election year, after controlling for month-of-year fixed effects and other covariates. This is a narrow within-year estimand, plausibly contaminated by year-end features—benchmark-cycle mechanics, holiday-related reporting differences, and year-end seasonal adjustment behavior—that are not election-specific in intent. The sign reversal indicates that an "election-year effect" is not robustly identified as a cross-cycle phenomenon.

Signed long-horizon revisions exhibit strong instability across time controls. Under trend and regime fixed effects, election-year months are associated with positive signed revisions (+101,000 to +106,000 jobs), strongly significant. Under year fixed effects, the estimate flips to negative (-58,000 jobs) and remains significant. This instability implies that directional inferences about election timing are highly sensitive to how time variation is absorbed. Sign reversal across defensible controls undermines any claim that the election-year indicator captures a stable directional bias.

Tail-risk results display similar fragility. At the 200,000-job threshold, trend and regime fixed effects yield large negative effects (-37 to -40 percentage points) with strong significance, while year fixed effects produce a positive effect (+12 percentage points), also significant. At the 100,000-job threshold, no significant effect emerges under trend or regime fixed effects, but year fixed effects yield a very large positive coefficient (+59 percentage points). At the 300,000-job threshold, no reliable pattern emerges in any specification, consistent with sparsity.

The all-cycles election-year evidence does not strengthen a manipulation hypothesis. Section 5.6.2 delivered a relatively clean result because the window is narrow and comparisons are primarily across months in a single late-sample regime. Section 5.6.3, by pooling election years, introduces substantial heterogeneity across macro conditions, benchmark-cycle environments, and institutional regimes. The sign flips under year fixed effects show that a pooled election-year indicator is not isolating a consistent timing phenomenon. There is no robust, cross-

Table 9: Election-Year Window Effects Across Outcomes

Outcome	Regime FE	Trend (t)	Year FE
abs_net_revision	-57.38^{**} (24.01)	-51.37^{**} (23.75)	$+62.55^{***}$ (16.85)
net_long_revision	$+100.95^{***}$ (34.82)	$+106.23^{***}$ (35.26)	-58.47^{**} (24.98)
net_pos	-0.121 (0.130)	-0.119 (0.128)	-0.655^{***} (0.134)
tail_100k	$+0.057$ (0.164)	$+0.088$ (0.166)	$+0.590^{***}$ (0.167)
tail_200k	-0.396^{***} (0.101)	-0.369^{***} (0.099)	$+0.121^{***}$ (0.045)
tail_300k	$+0.011$ (0.016)	$+0.029$ (0.023)	$+0.054$ (0.040)

Entries report the coefficient on ElectionYear_t, the election-year window indicator. Continuous outcomes measured in thousands of jobs. Binary outcomes are linear probability model coefficients. HAC standard errors (12 lags) in parentheses. All specifications include administration indicators, USREC, COVID shock, and month-of-year fixed effects. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

cycle election-year revision-risk signature that survives plausible time controls. Any apparent election-year effects are sensitive to whether the model identifies across years or within election years relative to December. The instability of coefficients across time controls—especially the systematic sign reversals under year fixed effects—implies that election-year indicators are confounded with within-year structure and broader time-varying factors.

5.7 Mechanism Probe: First-Print Magnitude and Revision Risk

Sections 5.3 through 5.6 establish that revision behavior exhibits timing-consistent structural breaks at COVID onset, stable conditioning on recession state, and fragile associations with political regime labels. A remaining question is whether a simple mechanical channel—large initial prints mechanically generate large revisions—accounts for revision risk once timing and regime structure are imposed. This section implements a falsification-oriented probe by augmenting the baseline specification with the absolute magnitude of the first employment estimate. The objective is not to validate a mechanism but to assess whether first-print size carries independent explanatory power after standard controls are applied.

For absolute long-horizon revision magnitude, the probe specification takes the form:

$$\text{abs\_net\_revision}_t = \beta_0 + \gamma' \text{Admin}_t + \beta_1 \text{USREC}_t + \beta_2 \text{COVID}_t + \beta_3 \left|\text{first\_estimate}_t\right| + \delta' \text{Month}_t + f(t) + \varepsilon_t,$$

where $|\text{first\_estimate}_t|$ denotes the absolute value of the first-release employment estimate, Month_t are month-of-year fixed effects, and $f(t)$ varies across linear trend, macro-regime fixed effects, and year fixed effects. The coefficient β_3 captures the conditional association between first-print magnitude and long-horizon revision magnitude, holding timing structure and regime controls fixed. Estimation uses OLS with HAC standard errors (12 lags).

The incremental fit from adding first-print magnitude is small. Across specifications, ΔR^2 ranges from 0.011 to 0.019, indicating that first-print size accounts for roughly one to two percent of residual variation in revision magnitude once baseline controls are included. The coefficient β_3 on $|\text{first\_estimate}_t|$ is negative across all three time controls, with point estimates ranging from -0.008 to -0.011. Statistical significance is marginal: p-values are approximately 0.06 under linear trend and year fixed effects, and 0.13 under regime fixed effects. The negative sign implies that larger first prints are conditionally associated with smaller absolute revisions, opposite the direction implied by a naive mechanical story in which large initial estimates are harder to correct and therefore generate larger revisions.

Table 10: First-Print Magnitude Probe Results

Specification	Baseline R^2	w/ First Print R^2	ΔR^2	β_3	SE	p-value
Trend (t)	0.240	0.254	0.013	-0.0086	0.0046	0.062
Regime FE	0.253	0.263	0.011	-0.0077	0.0052	0.134
Year FE	0.404	0.423	0.019	-0.0106	0.0057	0.062

Baseline model includes administration indicators, USREC, COVID shock, month-of-year fixed effects, and time controls. Augmented model adds $|\text{first\_estimate}_t|$. HAC standard errors (12 lags). $n = 300$.

The probe does not support the hypothesis that large first prints mechanically generate large revisions. The incremental explanatory power is small, the coefficient is negative rather than positive, and inference is sensitive to time controls. Even if the coefficient were stable and significant, it would remain a conditional association, unable to establish causal direction or discriminate among institutional channels affecting both first prints and later benchmarking. The correct interpretation is eliminative: simple size-driven revision risk stories do not carry independent weight once timing and regime structure documented in earlier sections are imposed.

5.8 Identification Boundaries and Limitations

Section 5 establishes which patterns appear timing-consistent, which are fragile under plausible time absorption, and which narratives fail basic robustness checks. Three findings emerge with confidence. First, COVID onset marks a sharp, sustained shift in long-horizon revision behavior that is timing-specific, measurement-channel-specific, and inference-robust. Second, recession state is a stable conditioning variable for both the magnitude and direction of payroll revisions. Third, presidential administration labels and election-year indicators do not form stable heterogeneity axes once reasonable time controls are imposed.

What these diagnostics do not establish is equally important. The interrupted time series design identifies a discontinuity but does not adjudicate among competing measurement disruption channels—response rate collapse, seasonal adjustment failure, establishment churn, or imputation protocol changes—that may have operated simultaneously. Recession coefficients reflect correlation with economic conditions, not recession-driven treatment effects. The absence of stable election-year patterns does not preclude the possibility of manipulation in specific instances; it indicates only that no systematic, cross-cycle signature survives plausible controls.

The diagnostic evidence presented here is not a substitute for forecasting evaluation. Whether revision risk can be characterized in real time using only information available at the time of first release requires a shift from retrospective pattern documentation to prospective uncertainty quantification. That reframing constitutes the subject of Section 6.

6. Predictive Modeling of Revision Risk

Building on the foundation of prior sections, Section 6 turns to predictive modeling of revision risk under strict real-time and deployability constraints. The objective is to evaluate whether long-horizon revision outcomes exhibit stable, out-of-sample regularities once models are restricted to information available at the time of the first payroll release and held to explicit feature-contract and leakage rules, rather than to construct an operational forecasting system. The analysis proceeds in eight steps. Section 6.1 formalizes admissible information sets, vintage constraints, and leakage controls, treating violations as grounds for invalidating downstream results. Section 6.2 defines the canonical modeling dataset, target variables, and evaluation philosophy, and Section 6.3 imposes a fixed temporal split structure that separates training, validation, COVID-era stress tests, and post-pandemic test periods. Section 6.4 constructs unconditional and trailing-window distributional baselines that serve as anchors for model evaluation. Sections 6.5 and 6.6 estimate and assess conditional quantile prediction models, focusing on interval calibration, coverage, and sharpness across slices and regimes. Section 6.7 introduces tail-risk warning diagnostics based on exceedance events, and Section 6.8 examines robustness, distributional shift, and failure modes under alternative training scopes and feature sets. Throughout, the organizing principle is diagnostic: predictive performance, calibration behavior, and breakdowns under shock are used to characterize the structure and limits of revision risk, not to validate deployable risk tools or overturn the identification boundaries established in Section 5.

6.1 Feature Contracts, Vintage Constraints, and Leakage Controls

Real-time prediction of revision risk is only as credible as the information set on which it is based. Before any model is estimated, it is necessary to formalize which variables are admissible as predictors at the time of the first payroll release, which variables are explicitly excluded, and how leakage controls are enforced. This subsection defines the feature contract governing all subsequent modeling, documents the associated leakage audits, and records the burn-in rule that determines the earliest month for which predictions are considered well-defined.

6.1.1 Admissible Information Set

The admissible information set comprises four feature groups, all constructed to respect strict vintage constraints. First, first-release level information includes the initial payroll estimate and its absolute value (`first_estimate`, `abs_first_estimate`), capturing the contemporaneous headline change available to users in real time. Second, calendar and time-index features consist of twelve month-of-year indicators and a time trend (`month_of_year_1-12`, `time_index`), allowing models to absorb residual seasonality and slow-moving shifts without relying on ex post regime labels. Third, lagged revision history is represented by one- through twelve-month lags of the absolute long-horizon revision (`abs_net_revision_lag1-12`) and rolling 12- and 24-month means and standard deviations (`abs_net_revision_roll_mean_12/24`, `abs_net_revision_roll_std_12/24`), each computed from strictly past data by applying a one-period shift before any rolling window is formed. Fourth, lagged tail-risk memory captures the recent frequency of extreme revisions via 12- and 24-month rolling tail-event rates (`tail_200k_freq_12`, `tail_200k_freq_24`), again constructed with a one-month shift followed by rolling aggregation.

These design choices are summarized in Table 11. All admissible predictors share two properties: they are functions of information observable at or before the current first release, and any use of multi-period history is implemented through lag and rolling operators that explicitly exclude contemporaneous outcomes. Under this contract, models are permitted to learn from past revision behavior, calendar timing, and the size of the current first print, but they cannot condition on information that would only become available after the initial release.

Table 11: Feature Contract Summary

Feature Group	Examples	Vintage Rule
First-release level	`first_estimate`, `abs_first_estimate`	Contemporaneous first print only
Calendar/time index	`month_of_year_1-12`, `time_index`	Deterministic calendar information at month t
Lagged revisions	`abs_net_revision_lag1-12`, `abs_net_revision_roll_mean_12/24`, `abs_net_revision_roll_std_12/24`	Past-only, with `shift(1)` applied before rolling windows
Lagged tail-risk memory	`tail_200k_freq_12`, `tail_200k_freq_24`	Past-only, with `shift(1)` + rolling exceedance frequency
Excluded diagnostics	`PAYEMS_CHG`, `USREC`, `covid_shock`	Not allowed as predictors; evaluation-only variables

6.1.2 Excluded Variables and Interpretation

The feature contract distinguishes sharply between deployable predictors and variables that are reserved for contextual interpretation or evaluation. Contemporaneous or ex post revision information is excluded from the predictor set: PAYEMS-based revision measures are tagged with feature group excluded, assigned `allowed = 0`, and annotated with the exclusion reason `contemporaneous_revision`, indicating that they capture information from later vintages rather than the initial release. Macroeconomic state indicators such as the NBER recession dummy `USREC` are likewise excluded and labeled as `ex_post_macro_indicator`, reflecting their dependence on business-cycle dating that is unavailable at the time of the first print. Finally, shock and regime indicators such as `covid_shock` are marked as `evaluation_only_indicator`, prohibiting their use as predictors while preserving them for slicing and stress-testing model performance across regimes.

Under this scheme, excluded variables play a strictly interpretive role. They are used to define evaluation slices, contextualize performance differences across economic states, and construct stress tests, but they never enter the admissible feature matrix. This separation prevents models from implicitly peeking at future information through labels that are themselves assigned ex post, and it ensures that any apparent predictability can be traced to genuinely real-time observables rather than to regime tags that embed hindsight.

6.1.3 Leakage Audits and Burn-In Rule

Vintage constraints are enforced through explicit leakage audits that operate independently of model specification. A `no_forward_looking_features` check verifies that all lagged and rolling predictors obey a one-period shift rule, confirming that rolling windows are applied only after shifting outcomes by at least one month and passing with no violations. A `no_contemporaneous_revision_info` check ensures that revision variables appear only in lagged or rolling form within the allowed feature set, with contemporaneous revision measures confined to the excluded group; this check also passes. A third audit, `shock_indicators_not_predictors`, confirms that COVID and related regime indicators are absent from the deployable feature set and used solely for evaluation purposes, again yielding a pass. Taken together, these audits operationalize the principle that any breach of vintage safety is treated as a hard failure rather than a tuning detail.

The interaction of lag and rolling windows implies a deterministic burn-in requirement that limits the effective sample. With a maximum lag length of 12 months and rolling windows of 12 and 24 months, the burn-in rule is summarized as

$$\max(\text{lag}, \text{rolling\_window}),$$

yielding an earliest usable month of 2002-01 once both lagged and 24-month rolling features are fully populated under the shift-plus-rolling protocol. Observations prior to this date are excluded from the modeling dataset, ensuring that all included months share a common, fully observed feature set. Under the feature contract defined here, any model that violates the burn-in rule or fails a leakage audit is deemed non-admissible, and its performance is not interpreted as evidence about the structure or predictability of payroll revisions.

6.2 Modeling Dataset Construction and Evaluation Design

The feature contract defined in Section 6.1 is implemented through a canonical modeling dataset that fixes both the target definitions and the timing structure for evaluation. This subsection documents the composition of that dataset, summarizes its missingness and feature availability, and formalizes the sample partitions used for training, tuning, and stress testing.

6.2.1 Dataset Structure and Targets

The modeling dataset is organized at a monthly frequency from 2002-01 through 2024-12, with each row corresponding to a first-release payroll month and its eventual long-horizon revision outcome. Two target variables are defined. The primary outcome y_t records the absolute long-horizon revision magnitude, while

$$y_{t,200k}^{\text{tail}} = \mathbf{1}\{y_t \geq 200{,}000\}$$

is a binary indicator equal to one when the absolute revision exceeds 200,000 jobs. Both targets are treated as non-features in the schema and are reserved exclusively for evaluation and loss computation.

Admissible predictors mirror the feature contract. Contemporaneous level information is captured by `first_estimate` and `abs_first_estimate`; calendar timing is encoded through twelve month-of-year dummies and an integer `time_index`; and dependence on past revision behavior is represented by twelve lags of the absolute revision (`abs_net_revision_lag1-12`), rolling 12- and 24-month means and standard deviations, and 12- and 24-month tail-event frequencies. Meta variables `month` and `slice` provide temporal ordering and evaluation partitioning but do not enter model fitting as features.

6.2.2 Missingness and Feature Availability

Schema and missingness diagnostics confirm that the dataset is fully consistent with the burn-in and vintage rules established earlier. All meta and target columns (`month`, `slice`, y_t, $y_{t,200k}^{\text{tail}}$) and all contemporaneous level, calendar, and time-index features exhibit zero missingness. Missing values arise only in lagged and rolling features and follow the mechanical pattern implied by the maximum lag and window lengths: the first twelve months show gradually increasing missing shares in `abs_net_revision_lag1-12`, rising from 0.36 percent for the first lag to 4.35 percent for the twelfth.

Rolling 12-month statistics and 12-month tail-event frequencies share the same 4.35 percent missing rate, reflecting the requirement of a full 12-month history under the `shift(1)` protocol. Rolling 24-month means, standard deviations, and 24-month tail frequencies display an 8.70 percent missing share, consistent with the need for 24 months of post-shift history. These patterns are entirely mechanical and are absorbed by the burn-in rule that restricts the effective modeling sample to months where all core features are defined. No ad hoc imputation or data-dependent filtering is applied beyond these deterministic constraints.

6.2.3 Evaluation Slices and Sample Partitions

Evaluation design is encoded directly in the `slice` column and summarized in Table 13. Each observation belongs to exactly one of four partitions that separate model fitting, tuning, and stress testing by calendar time. The `train` slice spans 2002-01 to 2016-12 and contains 180 observations, providing the pre-COVID history used for parameter estimation under strict past-only constraints. The `val` slice covers 2017-01 to 2019-12 with 36 observations and functions as a pre-shock holdout window for hyperparameter tuning and model selection.

The `stress_shock` slice isolates the COVID onset period from 2020-01 to 2020-12, with 12 observations reserved purely for evaluation of model behavior under an extreme but historically realized measurement shock. Finally, the `test_normal` slice extends from 2021-01 to 2024-12 with 48 observations, capturing post-pandemic conditions under which models trained on pre-shock data are evaluated for out-of-sample performance in a more stable environment.

Table 12: Evaluation Slices and Sample Size

Slice	Role	Start Month	End Month	Observations
`train`	Model fitting	2002-01	2016-12	180
`val`	Pre-shock tuning	2017-01	2019-12	36
`stress_shock`	COVID stress test	2020-01	2020-12	12
`test_normal`	Post-COVID test	2021-01	2024-12	48

Under this partitioning, all reported performance metrics in subsequent subsections are indexed by slice, ensuring that apparent improvements over baselines, calibration behavior, and failure modes are interpreted explicitly in relation to timing and regime rather than as undifferentiated summary statistics.

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is a binary indicator equal to one when the absolute revision exceeds 200,000 jobs. Both targets are treated as non-features in the schema and are reserved exclusively for evaluation and loss computation.

Admissible predictors mirror the feature contract. Contemporaneous level information is captured by `first_estimate` and `abs_first_estimate`; calendar timing is encoded through twelve month-of-year dummies and an integer `time_index`; and dependence on past revision behavior is represented by twelve lags of the absolute revision (`abs_net_revision_lag1-12`), rolling 12- and 24-month means and standard deviations, and 12- and 24-month tail-event frequencies. Meta variables `month` and `slice` provide temporal ordering and evaluation partitioning but do not enter model fitting as features.

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Schema and missingness diagnostics confirm that the dataset is fully consistent with the burn-in and vintage rules established earlier. All meta and target columns (`month`, `slice`, y_t, $y_{t,200k}^{\text{tail}}$) and all contemporaneous level, calendar, and time-index features exhibit zero missingness. Missing values arise only in lagged and rolling features and follow the mechanical pattern implied by the maximum lag and window lengths: the first twelve months show gradually increasing missing shares in `abs_net_revision_lag1-12`, rising from 0.36 percent for the first lag to 4.35 percent for the twelfth.

Rolling 12-month statistics and 12-month tail-event frequencies share the same 4.35 percent missing rate, reflecting the requirement of a full 12-month history under the `shift(1)` protocol. Rolling 24-month means, standard deviations, and 24-month tail frequencies display an 8.70 percent missing share, consistent with the need for 24 months of post-shift history. These patterns are entirely mechanical and are absorbed by the burn-in rule that restricts the effective modeling sample to months where all core features are defined. No ad hoc imputation or data-dependent filtering is applied beyond these deterministic constraints.

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Evaluation design is encoded directly in the `slice` column and summarized in Table 13. Each observation belongs to exactly one of four partitions that separate model fitting, tuning, and stress testing by calendar time. The `train` slice spans 2002-01 to 2016-12 and contains 180 observations, providing the pre-COVID history used for parameter estimation under strict past-only constraints. The `val` slice covers 2017-01 to 2019-12 with 36 observations and functions as a pre-shock holdout window for hyperparameter tuning and model selection.

The `stress_shock` slice isolates the COVID onset period from 2020-01 to 2020-12, with 12 observations reserved purely for evaluation of model behavior under an extreme but historically realized measurement shock. Finally, the `test_normal` slice extends from 2021-01 to 2024-12 with 48 observations, capturing post-pandemic conditions under which models trained on pre-shock data are evaluated for out-of-sample performance in a more stable environment.

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`test_normal`	Post-COVID test	2021-01	2024-12	48

Under this partitioning, all reported performance metrics in subsequent subsections are indexed by slice, ensuring that apparent improvements over baselines, calibration behavior, and failure modes are interpreted explicitly in relation to timing and regime rather than as undifferentiated summary statistics.

6.4 Distributional Baselines and Naïve Forecast Anchors

Before introducing conditional models, it is necessary to establish simple, transparent baselines that reflect the unconditional distribution of revision risk under the vintage constraints imposed in Section 6.1. This subsection defines two such baselines: global train-slice quantiles and trailing-window rolling quantiles. These serve as anchors against which more elaborate models are evaluated.

6.4.1 Global Quantile Baseline

The first baseline, `baseline_global_train_quantiles`, uses the empirical distribution of the primary target y_t on the `train` slice to construct constant prediction intervals. Specifically, the 0.10, 0.50, and 0.90 sample quantiles of

$$\{y_t : t \in \texttt{train}\}$$

are computed using only observations from 2002-01 to 2016-12, yielding values of approximately 9.9, 43.0, and 132.6 thousand jobs, respectively. These three quantiles

$$\hat{q}^{\text{train}}_{0.10}, \quad \hat{q}^{\text{train}}_{0.50}, \quad \hat{q}^{\text{train}}_{0.90}$$

define a fixed predictive band that is then applied unchanged to every month in the sample, regardless of slice, regime, or calendar time.

In the resulting baseline forecast file, each row records the realized outcome y_t alongside the triplet

$$\left(\hat{q}^{\text{train}}_{0.10}, \hat{q}^{\text{train}}_{0.50}, \hat{q}^{\text{train}}_{0.90}\right)$$

implied by this global distributional summary. The baseline is deliberately naive: it ignores all conditioning information, treats revision risk as stationary across time and regimes, and serves as a lower bound on what can be achieved by models that exploit calendar structure or lagged revision history. Any conditional model that fails to improve on the coverage and sharpness properties of this constant band offers limited evidence of meaningful structure in revision risk.

6.4.2 Trailing 24-Month Rolling Quantiles

The second baseline, `baseline_rolling24_quantiles`, introduces minimal temporal structure while preserving strict vintage safety. For each month t, the method computes trailing 24-month rolling 0.10, 0.50, and 0.90 quantiles of y_t using a past-only implementation:

$$\hat{q}_\tau(t) = \text{Quantile}_\tau \left(y_{t-1}, y_{t-2}, \ldots, y_{t-24} \right), \qquad \tau \in \{0.10, 0.50, 0.90\}.$$

Operationally, this corresponds to `y.shift(1).rolling(24)`, so that only information available up to the prior month enters the estimate. The resulting values $\hat{q}_\tau(t)$ vary over time, adapting gradually to changes in the observed revision distribution while remaining agnostic about calendar effects, macro regimes, or lagged predictors beyond the target itself.

This rolling baseline captures slow-moving changes in revision risk, such as extended high-volatility regimes, without invoking an explicit model for conditional dependence on features. It is particularly informative for evaluating whether more sophisticated quantile models add value beyond simply tracking local unconditional behavior. If a conditional model does not meaningfully tighten intervals or improve coverage relative to this trailing-window benchmark, especially in the `stress_shock` and `test_normal` slices, its additional complexity is unlikely to reflect stable, exploitable structure in the revision-generating process.

6.5 Conditional Quantile Prediction Models

Having established distributional and trailing-window baselines, attention turns to conditional quantile models that attempt to exploit the admissible feature set defined in Section

6.1. The objective is to assess whether absolute long-horizon revision risk exhibits stable conditional structure once models are constrained to real-time information and evaluated under the temporal design of Sections 6.2–6.3, rather than to construct deployable risk forecasts.

6.5.1 Specification and Estimation Framework

Conditional quantile models are implemented as linear quantile regressions estimated on the modeling dataset with the feature contract and temporal splits described earlier. For each quantile level $\tau \in \{0.10, 0.50, 0.90\}$, the model solves

$$\min_{\beta_\tau} \sum_t \rho_\tau \left(y_t - x_t^\top \beta_\tau \right), \qquad \rho_\tau(u) = u \left(\tau - \mathbf{1}\{u < 0\} \right),$$

and delivers conditional quantile predictions

$$\hat{q}_\tau(t) = x_t^\top \hat{\beta}_\tau.$$

Two model variants are considered. The first, `qreg_linear_exclshock`, trains through 2019-12 and excludes COVID-era observations from the training window. The second, `qreg_linear_inclshock`, extends the training window through 2020-12, allowing the model to learn from the initial COVID shock while preserving the same feature set and evaluation slices. In both cases, regressors comprise all allowed predictors except the month-of-year dummies, which remain excluded from the quantile feature set to avoid redundancy with calendar indicators elsewhere in the modeling pipeline.

Quantile levels are fixed at 0.10, 0.50, and 0.90, corresponding to lower, median, and upper conditional quantiles of y_t. A shared tuning step selects regularization strength via cross-validation confined to the pre-shock period: the preferred penalty levels are $\alpha = 0.1$ for the 0.10 quantile, $\alpha = 0.5$ for the median, and $\alpha = 0.05$ for the 0.90 quantile, chosen to balance interval sharpness and stability across the three folds defined in Section 6.3. Convergence logs document successful optimization for all quantiles and both training scopes, with no failed fits or degenerate solutions.

6.5.2 Monotonicity Enforcement and Prediction Outputs

Because separate quantile regressions are estimated for each τ, the raw predictions need not respect the logical ordering

$$\hat{q}_{0.10}(t) \leq \hat{q}_{0.50}(t) \leq \hat{q}_{0.90}(t)$$

at every month. A post-estimation monotonicity pass therefore scans predicted triplets and applies minimal adjustments where necessary. The monotonicity log shows that no corrections are required in the early sample; the first recorded fixes appear in 2004-01, where the original ordering

$$(\hat{q}_{0.10}(t), \hat{q}_{0.50}(t), \hat{q}_{0.90}(t))$$

is overwritten by the sorted triplet

$$(\text{min}, \text{median}, \text{max})$$

to restore non-crossing intervals. Similar adjustments occur sporadically throughout the sample for both `qreg_linear_exclshock` and `qreg_linear_inclshock`, but in each case the final ordering preserves the original values up to sorting rather than re-estimating coefficients.

The prediction file aggregates these outputs across models and slices. For each month t, it records the realized outcome y_t, the model identifier, the slice label, and the adjusted conditional quantiles $\hat{q}_{0.10}(t)$, $\hat{q}_{0.50}(t)$, and $\hat{q}_{0.90}(t)$. In the pre-shock `train` and `val` slices, quantile bands remain relatively narrow and track the dispersion of revisions documented in Section 4, with upper quantiles typically in the 100–140 thousand range and lower quantiles near 10 thousand. In the `stress_shock` slice, conditional upper quantiles for `qreg_linear_exclshock` rise into the 180–230 thousand range during 2020, widening further when the model is trained including shock months, consistent with the elevated volatility and tail exposure identified in the quasi-causal diagnostics.

6.5.3 Interpretation and Limits

These conditional quantile models provide a disciplined way to summarize how revision risk varies with first-release levels, calendar timing, and lagged revision history under strict vintage constraints. However, their predictive content is interpreted narrowly. First, the linear specification and modest gains over the global and rolling baselines indicate that much of the revision environment remains dominated by heavy tails and regime shifts rather than by smooth, feature-driven variation. Second, the need for monotonicity fixes at various points in the sample underscores that even well-tuned quantile regressions can produce locally incoherent intervals in fat-tailed settings, particularly when the underlying distribution shifts abruptly.

Finally, the contrast between `qreg_linear_exclshock` and `qreg_linear_inclshock` is used diagnostically rather than to justify training on shock periods. Models trained including COVID months tend to produce wider upper quantiles in the `test_normal` slice, reflecting a form of variance scar that persists even after the immediate crisis subsides. This behavior is consistent with the quasi-causal evidence that tail-risk probabilities remain structurally elevated post-COVID and reinforces the interpretation that predictive modeling can describe, but not fully tame, the instability of payroll revisions documented in Sections 4 and 5.

6.6 Calibration, Coverage, and Interval Behavior

Section 6.5 established that conditional quantile models can exploit real-time features to produce non-crossing prediction intervals for absolute long-horizon revisions, but left open how well those intervals behave in practice relative to simple distributional baselines. This subsection evaluates calibration, coverage, and width for 10–90 percent intervals across slices and through time, using the global and rolling baselines as reference points and treating departures from nominal behavior as evidence about the limits of probabilistic prediction in this environment.

6.6.1 Quantile Calibration Across Slices

Slice-level calibration diagnostics summarize how often realized outcomes fall below the 0.10, 0.50, and 0.90 quantiles for each method and evaluation window. For the global train-slice baseline, empirical hit rates in the 2021–2024 `test_normal` period are approximately 0.02 for $q_{0.10}$, 0.27 for $q_{0.50}$, and 0.69 for $q_{0.90}$, indicating severe under-use of the lower quantile and moderate under-coverage at the upper quantile relative to the nominal $(0.10, 0.50, 0.90)$ targets. The rolling 24-month baseline performs better in normal conditions, with test hit rates near 0.17, 0.56, and 0.90 for $(q_{0.10}, q_{0.50}, q_{0.90})$, but calibration degrades sharply in the 2020 `stress_shock` slice, where the upper quantile hit rate falls to about 0.50 despite substantially widened intervals.

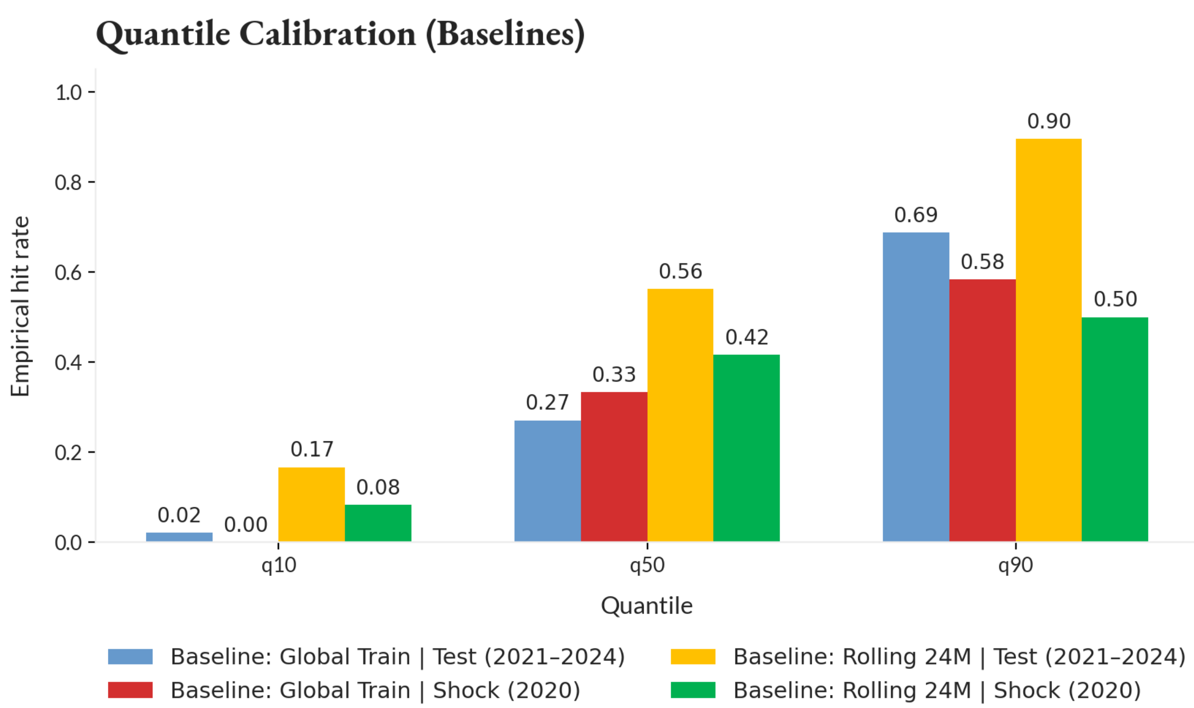

Figure 12: Quantile calibration for global and rolling baselines. Bars report empirical hit rates by quantile and slice. The global baseline under-covers the upper tail in normal periods, while the rolling 24-month baseline improves calibration but widens substantially during the 2020 stress period.

Quantile regression models improve median and upper-tail calibration while matching the baselines' poor lower-tail behavior. In the `test_normal` slice, both `qreg_linear_exclshock` and `qreg_linear_inclshock` achieve hit rates around 0.27 for $q_{0.50}$ and 0.79–0.83 for $q_{0.90}$, bringing the upper tail close to nominal without inflating intervals to the extent of the rolling baseline. However, the 0.10 quantile remains effectively unused: empirical hit rates are roughly 0.02 in 2021–2024 and zero in 2020 for all four methods, indicating that none of the approaches reliably anticipates unusually small revisions.

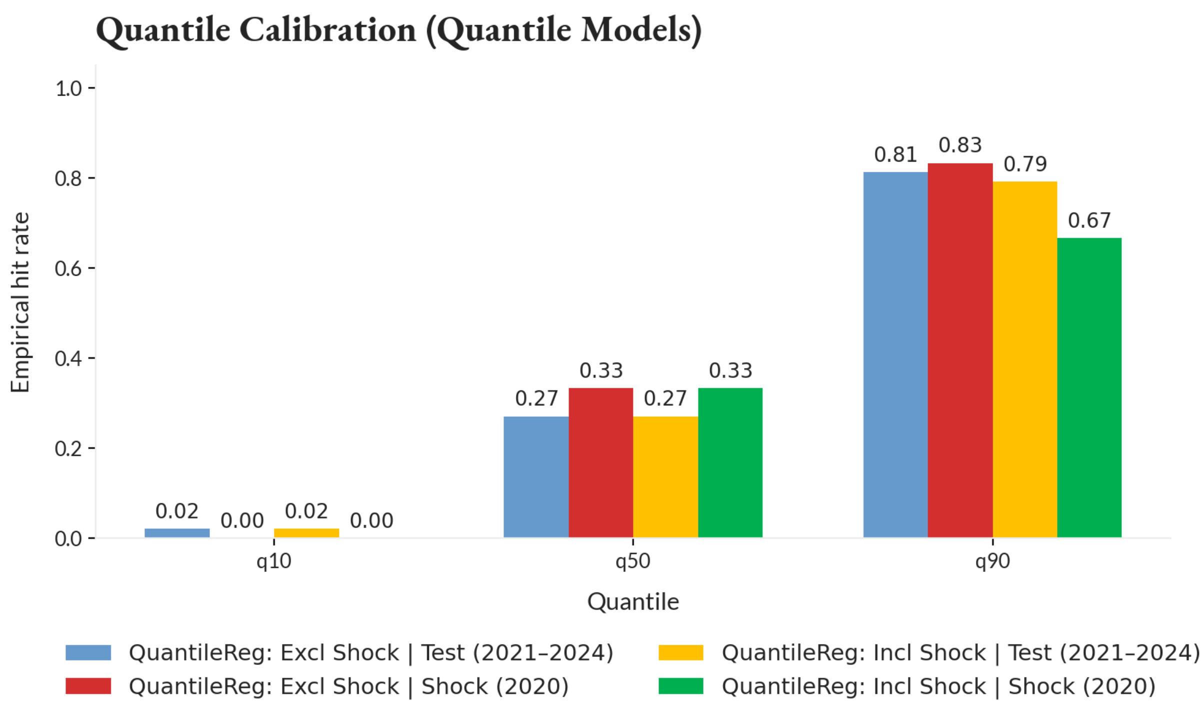

Figure 13: Quantile calibration for conditional quantile regression models. Both models achieve improved upper-tail calibration in the `test_normal` slice relative to the global baseline, while avoiding the most extreme width inflation observed in the rolling baseline during the stress period.

6.6.2 Interval Coverage and Width Over Time

Time-series diagnostics complement slice-level calibration by tracking 10–90 coverage and width through the full sample. The interval time-series file records, for each method and month, whether the realized revision falls inside the 10–90 interval and the associated width; these series are then smoothed using rolling means to highlight regime shifts. For the global baseline, coverage oscillates between roughly 0.3 and 1.0 across the business cycle despite a constant width of approximately 123 thousand jobs, with prolonged under-coverage during the 2008–2009 financial crisis and the 2020 COVID shock. The rolling baseline exhibits more adaptive coverage, but coverage falls well below 0.5 at the height of both crises, confirming that even very wide intervals fail to accommodate the most extreme revisions.

The quantile regression models display similar qualitative behavior. In expansion periods, rolling coverage for `qreg_linear_exclshock` and `qreg_linear_inclshock` typically lies between 0.8 and 1.0, broadly consistent with nominal 10–90 intervals, while interval widths cluster in the 100–130 thousand range, slightly above the global baseline but well below the rolling baseline. During 2008–2009 and 2020, coverage plunges to levels comparable with or only slightly better than those of the baselines, even as widths expand into the 160–200 thousand range, illustrating that conditional structure cannot fully absorb regime breaks of the magnitude documented in Section 5.

6.6.3 Slice-Level Summary and Interpretation

Slice-level scoreboard metrics summarize the trade-off between calibration and sharpness. On the 2021–2024 `test_normal` slice, both quantile models achieve 10–90 coverage of approximately 0.77–0.79, compared with 0.67 for the global baseline and 0.73 for the rolling baseline. Mean interval widths over the same period are roughly 143–148 thousand jobs for the quantile regressions, modestly wider than the global baseline's 123 thousand but substantially narrower than the rolling baseline's 233 thousand. Pinball losses are consistently lower for the conditional models across all three quantiles, indicating improved distributional fit relative to both unconditional anchors.

In the 2020 `stress_shock` slice, coverage deteriorates for all methods. The global baseline covers approximately 0.58 of realizations inside its 10–90 band. The rolling baseline reaches roughly 0.42 coverage despite average widths exceeding 190 thousand jobs. The quantile models fall between these extremes, with coverage in the 0.66–0.83 range and mean widths near 168–192 thousand. These patterns indicate that conditional models deliver modest improvements in

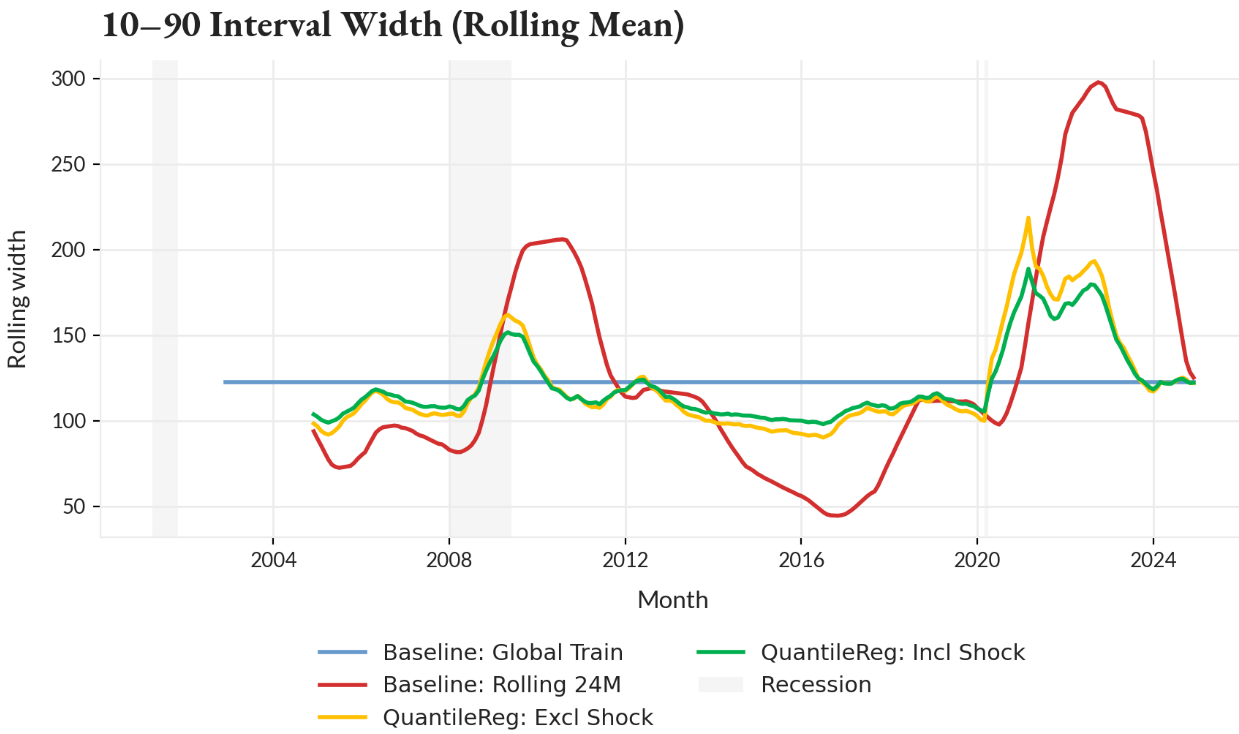

Figure 14: Rolling 10–90 interval width over time for global and rolling baselines and conditional quantile models. The global baseline maintains a constant band, the rolling baseline expands sharply during crisis periods, and the quantile regression models display intermediate responsiveness, widening under stress without permanently adopting crisis-level dispersion.

calibration and sharpness during normal regimes but do not neutralize the structural tail amplification documented in Section 5. Under extreme shocks, large revisions remain systematically under-predicted regardless of specification.

6.7 Tail-Risk Warning Diagnostics

The quantile models in Section 6.5 summarize revision uncertainty through 10–90 percent intervals but do not directly target exceedances of large thresholds such as 200,000 jobs. This subsection introduces explicit tail-risk warning scores that approximate the probability of a 200k exceedance and evaluates their behavior across regimes, with particular attention to misses of realized extremes. Tail diagnostics are used purely as descriptive tools for the structure of tail events; they are not interpreted as deployable early-warning systems.

6.7.1 Tail-Event GLM Specification

Tail warnings are constructed using logistic generalized linear models that predict the binary indicator $y^{\text{tail}}_{t,200k}$ defined in Section 6.2, equal to one when the absolute long-horizon revision exceeds 200,000 jobs. For each month t, the GLM models

$$p_t = \Pr\left(y^{\text{tail}}_{t,200k} = 1 \mid x_t\right) = \sigma(x_t^\top \gamma), \qquad \sigma(z) = \frac{1}{1+e^{-z}},$$

with parameters γ estimated on the pre-specified training window.

Two specifications mirror the quantile-regression runs. Model `glmtail200k_exclshock` trains through 2019–12, excluding COVID months from the estimation sample, while `glmtail200k_inclshock` extends the training window through 2020–12 and thus incorporates the COVID shock itself. Both models use an ℓ_1-penalized logistic link with penalty parameter $C = 0.05$, selected via pre-2017 cross-validation to balance sparsity and predictive performance. Both rely on the same admissible feature set used for quantile regression, preserving the feature contract and vintage rules.

The resulting prediction file records, for each month and model, the realized tail indicator $y^{\text{tail}}_{t,200k}$, the GLM-implied exceedance probability p_t, and the evaluation slice. Slice-level scoreboards indicate that, on the 2021–2024 `test_normal` slice, both GLMs achieve precision around 0.17–0.20 and recall near 0.625 at operating points calibrated on the pre-2017 period, yielding precision–recall AUCs of approximately 0.63 for both the excl- and incl-shock models. In the `stress_shock` slice, precision falls to about 0.13–0.14 with recall fixed at 0.5 and PR-AUCs dropping to roughly 0.25–0.27, highlighting the difficulty of maintaining discriminating tail warnings during COVID-era extremes.

A related scoring rule, used in the slice-level scoreboards, is the Brier score for a slice S,

$$\text{Brier}(S) = \frac{1}{|S|} \sum_{t \in S} \left(y^{\text{tail}}_{t,200k} - p_t\right)^2,$$

which summarizes the mean squared error of the probability forecasts. Lower Brier scores on the `train` and `test_normal` slices for the excl-shock specification reflect better overall calibration in those regimes relative to the incl-shock variant.

6.7.2 Reliability of Tail Warnings

Reliability tables assess how well GLM warning scores track empirical tail frequencies in different probability bands. For the 2021-2024 `test_normal` slice, selected bins from the reliability table are summarized in Table 14.

Table 14: GLM Tail-Warning Reliability (test_normal, 200k Threshold)

Model	Probability Bin	Avg. Score in Bin	Event Rate
`glmtail200k_exclshock`	0.2-0.33	0.28	0.33
`glmtail200k_exclshock`	0.8-0.95	0.85	0.20
`glmtail200k_exclshock`	0.9-1.0	0.96	0.67
`glmtail200k_inclshock`	0.4-0.51	0.45	0.17
`glmtail200k_inclshock`	0.8-0.95	0.84	0.60
`glmtail200k_inclshock`	0.9-1.0	0.92	1.00

Entries report average predicted probabilities and realized tail-event frequencies within selected probability bins for the 2021-2024 `test_normal` slice. Tail events are defined as absolute long-horizon revisions exceeding 200,000 jobs.

As Table 14 shows, higher probability bins are associated with higher realized tail frequencies, particularly for `glmtail200k_inclshock`, where scores above 0.8 correspond to event rates between 0.60 and 1.0. At the same time, mid-range bins, for example probabilities around 0.28-0.45 with event rates near 0.17-0.33, reveal that calibration is coarse and noisy. This reflects both the rarity of 200k events and the limited ability of linear GLMs to track subtle shifts in tail risk.

Calibration deteriorates in the 2020 `stress_shock` slice for both models. For `glmtail200k_exclshock`, probability bins around 0.27 and 0.31 show event rates of 0.33 and 0.0, respectively, while the 0.8-0.92 bin exhibits an event rate of 0.5 and the highest 0.9-1.0 bin records no events at all. The `glmtail200k_inclshock` model displays a similar pattern, with an event rate of about 0.2 in the 0.4-0.55 bin, 0.5 in the 0.8-0.92 bin, and 0.0 in the 0.9-1.0 bin. These reliability results reinforce that GLM warnings can rank-order risk to some extent in stable periods but do not deliver fully calibrated tail probabilities, especially during regime breaks.

6.7.3 Tail Misses Relative to $q_{0.90}$ Intervals

To connect tail warnings to the interval framework in Section 6.6, a tail miss diagnostic compares realized absolute revisions to the predicted 90th percentile $\hat{q}_{0.90}(t)$ from the quantile regression models. For each month, the diagnostic table records the realized revision y_t, the predicted upper quantile $\hat{q}_{0.90}(t)$, and an indicator for whether the revision exceeds $\hat{q}_{0.90}(t)$. Exceedances correspond to intervals that fail to cover the realized outcome at the upper tail. The resulting time series shows clusters of $q_{0.90}$ exceedances around the 2008-2009 financial crisis and, more prominently, the 2020-2022 COVID and post-COVID period, mirroring the heavy-tail and regime-shift patterns documented earlier.

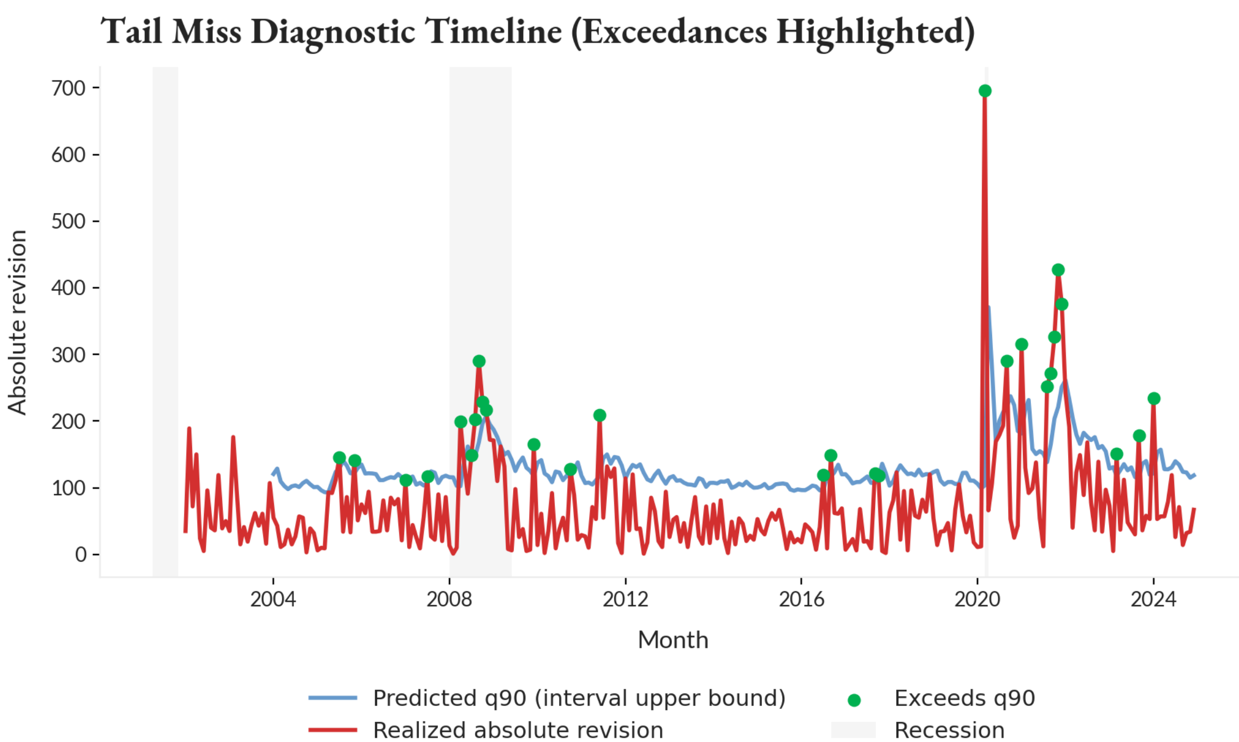

Figure 15: Tail miss diagnostic timeline. The realized absolute long-horizon revision is plotted against the predicted upper quantile $\hat{q}_{0.90}(t)$ from the conditional quantile model. Months in which $y_t > \hat{q}_{0.90}(t)$ are highlighted as upper-tail exceedances. Clusters of exceedances occur during the 2008-2009 financial crisis and the 2020-2022 COVID and post-COVID period, indicating repeated interval under-coverage during structural shifts.

Many of the largest tail events, especially the extreme COVID-era revisions, lie well above the predicted $q_{0.90}$ even when GLM warning scores are elevated. This indicates that joint calibration of interval upper bounds and exceedance probabilities remains imperfect. Viewed together with the precision-recall and reliability diagnostics, the evidence suggests that real-time features support useful but fragile tail warnings. They can flag elevated risk and improve ranking of high-risk months in normal regimes, yet they still miss a substantial fraction of the largest realized revisions, particularly when the revision process undergoes abrupt structural change.

7. Discussion and Policy Implications

Empirical and diagnostic evidence presented in the preceding sections establishes that payroll revisions are not merely random noise diminishing over time, but structural features of a measurement system that encounters specific limits during periods of economic transition. By characterizing the distribution of revisions in Section 4, isolating the timing and persistence of shock-induced volatility in Section 5, and documenting the breakdown of predictive calibration during stress events in Section 6, this analysis supports a reinterpretation of how preliminary labor market data should be utilized in real-time decision-making. This section synthesizes those findings to address broader implications for measurement credibility, interpretation of early signals, and the structural nature of revision risk.

7.1 The Structural Limits of Real-Time Measurement

Failure of even sophisticated conditional quantile models to reliably capture lower-tail risks during the COVID-19 onset underscores a fundamental limitation: the informational content required to identify a turning point is often absent from the initial sample entirely. This limitation is not a failure of modeling but of measurement architecture. The Current Employment Statistics program relies on a voluntary sample that is naturally subjected to friction during periods of rapid establishment churn. Crucially, the Net Birth/Death Model, used to impute employment from new business formation, depends on ARIMA-based projections of historical patterns. By construction, this methodology assumes that past relationships between business formation and employment growth will persist into the current period. During cyclical turning points or exogenous shocks, this assumption breaks down precisely when accuracy is most needed.

Burn-in constraints and the inability of vintage-based features to anticipate the magnitude of the 2020 collapse demonstrate that the initial signal in such environments is not simply a noisy version of the truth; it is, for a brief window, a qualitatively different object than the final benchmark. Policy reliance on the point estimate of the first print during such windows implicitly assumes stability of the revision process that the data do not support. Recognizing this structural blind spot requires treating early estimates not as definitive counts but as provisional signals whose confidence intervals must expand nonlinearly when auxiliary indicators suggest regime transition.

The lag structure of administrative data compounds this blindness. The Quarterly Census of Employment and Wages, which serves as the benchmark for revisions, becomes available only with a lag of five to nine months. This creates an unavoidable informational gap in which real-time decisions must be made on survey data lacking comprehensive tax-record coverage. Machine learning models trained on past revisions can quantify the probability of large adjustments, but they cannot close this gap because the feature space available in real time does not contain the target signal until later vintages arrive.

7.2 Regime-Dependent Uncertainty and Monetary Policy

Standard practice in financial markets and policy analysis often involves applying a static rule of thumb for revision risk, typically assuming that a first print is accurate within a fixed margin such as plus or minus 60,000 jobs. Findings in Sections 4.3 and 5.3 render this static approach indefensible. The timing-consistent discontinuity in revision behavior triggered by the COVID-19 pandemic, coupled with the robust association between recessions and downward revision bias, establishes that reliability is state dependent.

Implications for the Federal Reserve's data dependence mandate follow directly from this regime sensitivity. If monetary policy is calibrated to incoming data, but the reliability of that data varies systematically with economic regime, then a policy stance based on initial prints risks becoming procyclical. During the onset of a recession, first prints tend to overstate employment strength and are revised downward later. Central banks reacting to strong initial payrolls may tighten or hold rates too long, unaware that underlying conditions are already deteriorating. Evidence of regime-dependent signal-to-noise deterioration implies that data dependence must be qualified by regime awareness.

An evidence-based approach to data consumption would therefore abandon static confidence intervals in favor of a regime-contingent framework, effectively a red, yellow, green reliability classification. In a benign expansionary regime coded green, revision dispersion is contained and the first print is highly informative. In a high-variance or recessionary regime coded red, that same signal carries a materially different noise-to-signal ratio, with elevated probability of large, negative, and heavy-tailed adjustments. Policymakers should explicitly weight incoming data less heavily when internal volatility metrics, such as the tail-risk probabilities modeled in Section 6.7, indicate a high-variance environment.

7.3 Institutional Credibility Versus Political Narrative

The magnitude of revisions observed in 2024, which motivated this study, has frequently been framed through a political lens. Quasi-causal diagnostics in Section 5 suggest a different interpretation. The absence of robust administration-specific effects, combined with clear identification of a persistent volatility scar following the COVID-19 shock, indicates that recent revision episodes are consistent with institutional stress rather than political manipulation.

The post-2020 labor market appears characterized by a permanently higher baseline of measurement difficulty, likely driven by structural changes in work arrangements, business formation dynamics, and response patterns that have not fully reverted to pre-pandemic norms.

Large revisions in this context are symptoms of a measurement apparatus confronting a more complex economy, not evidence of partisan bias. This distinction is critical for maintaining the public trust function of statistical agencies. When large revisions occur, presenting the initial number as definitive invites skepticism. Communicating uncertainty more transparently, potentially by publishing a real-time revision risk score alongside the headline number, could strengthen long-term credibility during high-stress periods.

7.4 Informational Asymmetry and Market Efficiency

The temporal structure of revisions and the persistence of tail risk documented in Sections 4.5 and 6.7 carry implications for market efficiency. The weak correlation between short-horizon revisions from the first to the third print and the eventual long-horizon benchmark outcome implies that intermediate updates often fail to converge smoothly toward the administrative truth.

This disconnect creates a prolonged period of informational asymmetry in which capital allocation and monetary policy decisions are made based on provisional data that may not converge to benchmark reality for years. The heavy-tailed nature of this divergence means that eventual corrections are frequently abrupt rather than gradual. During suspected regime change, the cost of overreacting to a single monthly print becomes asymmetric because the likelihood of substantial revision is highest precisely when economic conditions are unstable. Incorporating revision-risk models into valuation and macro frameworks may therefore mitigate exposure to noise-driven mispricing.

7.5 Recommendations for Statistical Agencies

The limited predictive power of internal revision history documented in Section 6 suggests diminishing returns from exclusive reliance on survey-based methods and historical imputation. To better capture turning points, statistical agencies should accelerate integration of high-frequency private data, such as aggregated payroll processor records or financial transaction flows, into first-print estimation procedures.

Private data sources, while imperfect, do not suffer from identical response lag or birth-death blind spots as the official survey. A hybrid estimation framework that weights survey responses against real-time private flows could provide a more robust signal during transition periods when administrative benchmarks have not yet materialized. Such an approach would shift the objective from measuring the survey to nowcasting the eventual benchmark, aligning preliminary releases more closely with underlying economic reality.

8. Conclusion

This study has systematically evaluated the reliability, distributional structure, and predictability of U.S. nonfarm payroll revisions over the 2000–2024 period. By constructing a vintage-consistent dataset and applying a sequence of exploratory, quasi-causal, and machine-learning diagnostics, the analysis challenges the conventional view of revisions as benign, mean-zero statistical noise. Instead, the evidence establishes that measurement error in the labor market is a heavy-tailed, regime-dependent process that undergoes structural shifts precisely when accurate real-time information is most critical.

8.1 Summary of Contributions

The primary contribution of this research is the formal characterization of structural measurement risk across regimes and vintages. The distributional analysis in Section 4 demonstrated that while typical revisions are small, the revision process is dominated by heavy tails that defy Gaussian assumptions. Section 5 provided quasi-causal evidence that the COVID-19 pandemic induced a persistent volatility scar on the measurement apparatus, creating a distinct high-variance regime that has not fully reverted to pre-2020 baselines. The predictive modeling in Section 6 revealed the limits of internal history: while machine learning models can identify periods of elevated risk, they systematically fail to calibrate correctly during shock onsets, confirming that the informational content required to predict a turning point is often structurally absent from the initial data vintage.

These findings argue for a shift in how preliminary data are consumed. The first print should not be interpreted as a point estimate with a static margin of error, but as a regime-contingent signal whose reliability degrades nonlinearly during economic stress. The identification of turning-point bias, the tendency for initial estimates to overstate employment at the onset of contractions, provides an empirically grounded caution for policymakers relying on real-time data to calibrate monetary or fiscal interventions.

8.2 Limitations

While this study offers a comprehensive diagnostic of revision behavior, it does not propose a deployable solution for nowcasting the final benchmark. The machine learning models developed here are diagnostic tools intended to bound uncertainty, not operational forecasting systems capable of replacing the Bureau of Labor Statistics methodology. The analysis is restricted to the Establishment Survey and its specific benchmarking protocols; it does not evaluate whether similar regime-dependent fragility exists in the Household Survey or other administrative series. The identification of the COVID-19 structural break relies on a single, albeit large, event. Distinguishing this specific institutional shock from generic cyclical volatility remains a challenge inherent to macro-empirical work with a limited number of recession episodes in the sample.

8.3 Future Directions

The structural limits identified in this analysis suggest that future improvements in real-time accuracy will likely require moving beyond the current survey-based paradigm. Future research should investigate the integration of high-frequency private data, such as real-time payroll processor records or financial transaction flows, directly into the initial estimation process. A hybrid modeling framework that weights official survey returns against these external, census-like signals could potentially narrow the informational gap during the months before administrative benchmarks become available. Extending the regime-dependent reliability framework to sectoral levels could isolate which industries drive the heavy-tailed behavior of aggregate revisions, allowing for more targeted risk adjustments in real-time analysis.

Ultimately, while the demand for immediate economic data is immutable, interpretation of that data must evolve in light of the evidence presented here. By acknowledging the structural, state-dependent nature of measurement uncertainty, policymakers and market participants can better navigate the gap between the economy as initially reported and the economy as later benchmarked.

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